



HISTOLOGY OF EPITHELIA

Thomas F. Freddo, O.D., Ph.D.
Professor
MCP Health Sciences University

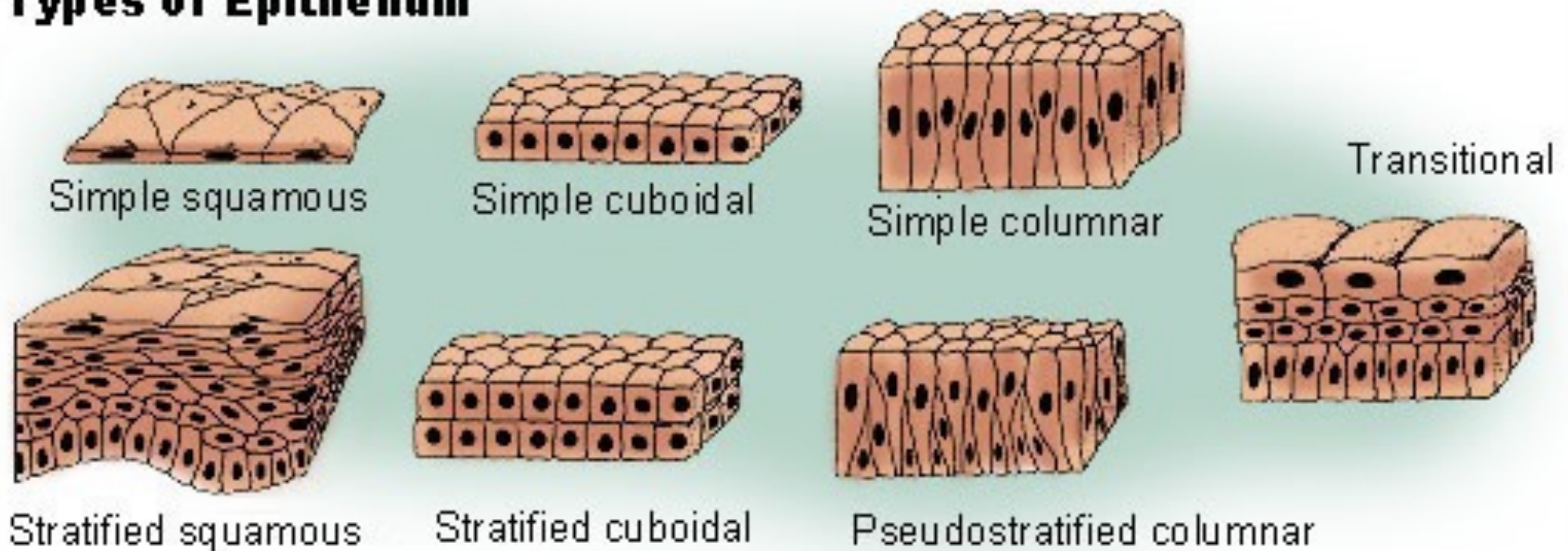
FUNCTIONS OF EPITHELIA

- The principal function of an epithelium is to separate two different environments while allowing those two environments to interact in a controlled fashion.
- In most cases, an epithelium separates a surface (internal or external) from an underlying connective tissue space.
- Interactions can occur through:
 - Selective Secretion
 - Selective Absorption
 - Transcellular transport
 - Picking up messages (chemical, sensorial) and conveying them to other systems for a response.
- Epithelial tissues are polarized. They have an apical surface facing the extracellular environment and basal surface that elaborates a basement membrane that binds that epithelium to an underlying connective tissue.
- Epithelia invariably contain sensory nerves (in some cases, like cornea, these nerves provide a necessary trophic effect)
- Epithelial layers do not generally contain blood vessels.

TYPES OF EPITHELIUM

- **Simple** – a single layer of polarized cells resting on a basement membrane
- **Stratified (compound)** – multiple layers that all differentiate from a basal layer
- **Pseudostratified** – a single layer of cells of various heights
- **Transitional** – an epithelium that changes between two morphological categories with repeating functional cycles.
- **Myoepithelium** – A cell exhibiting an epithelial portion (usually toward the apical surface) and a modified smooth muscle cell-like portion (usually toward the basal surface).

Types of Epithelium



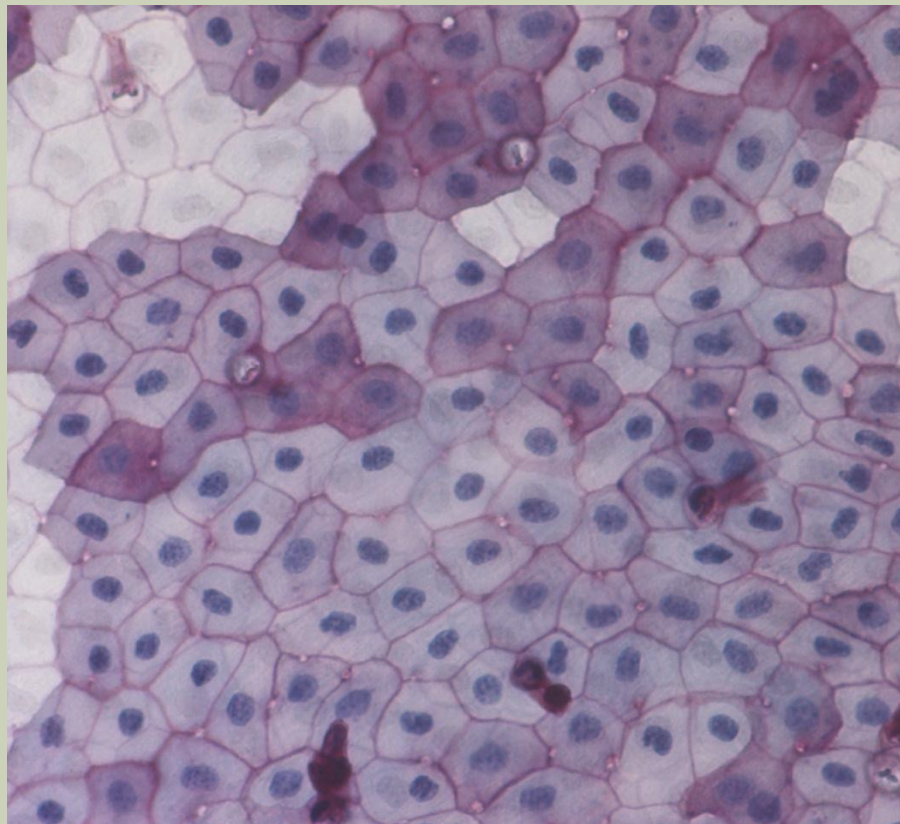
SIMPLE SQUAMOUS EPITHELIUM

Single layer of flattened, polarized cells resting on a basement membrane

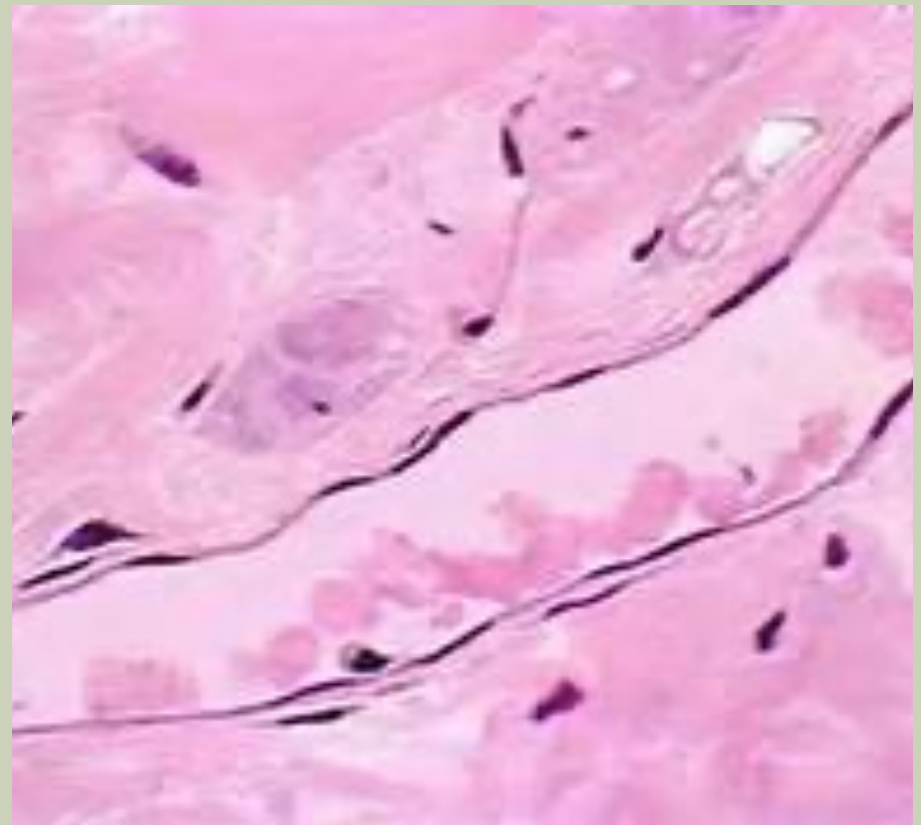
In en-face view, normal healthy squamous epithelial cells have a roughly hexagonal shape.

In Cross-section, the nuclei are thin and elongated (fusiform).

En-face



Cross-section



SIMPLE SQUAMOUS EPITHELIUM

Examples Throughout the Body

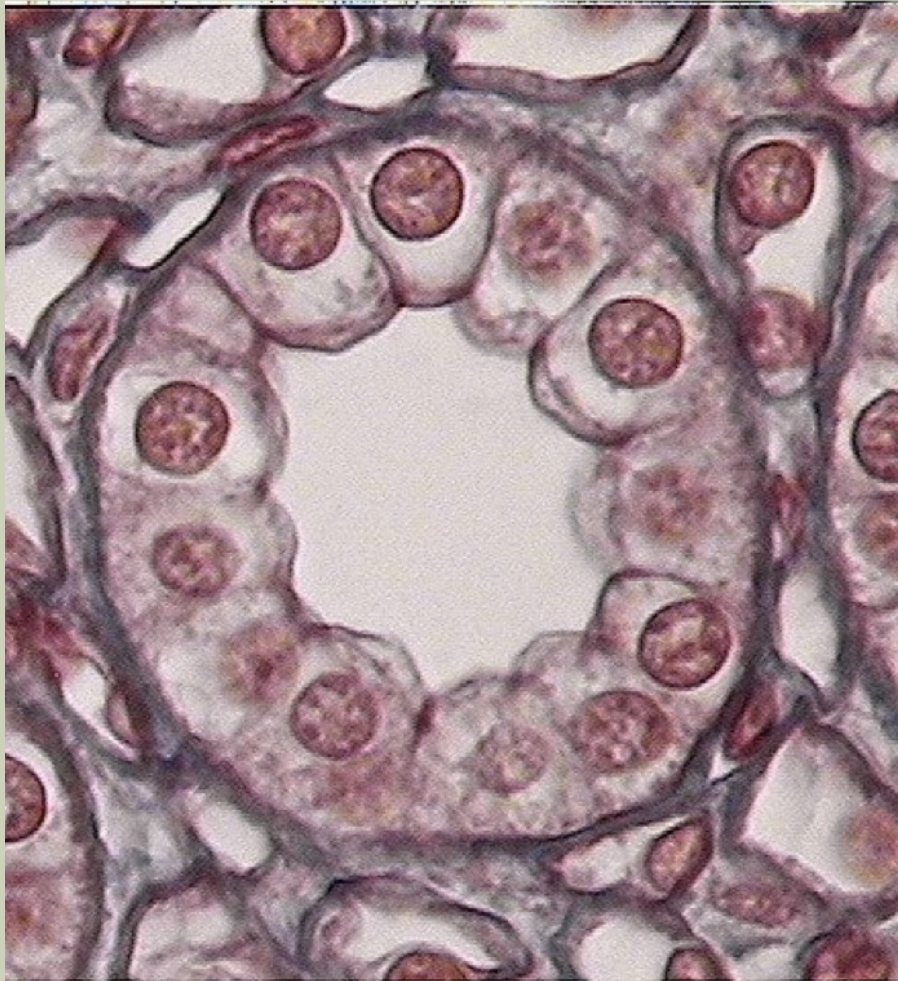
- Vascular endothelial cells
- Lymphatic endothelial cells
- Pulmonary epithelium
- Glomerulus of the kidney

Examples in the Eye

- Corneal endothelium
- Trabecular endothelium
- Inner and outer wall of Schlemm's canal
- Vascular endothelial cells

They tend to occur where there is rapid exchange of fluid or fluid soluble ions and gases.

SIMPLE CUBOIDAL EPITHELIUM



- Single layer of polarized cells resting on a basement membrane
- Cells are box-like, with similar height and width
- In cross-section, the nuclei are round and centrally-placed.

SIMPLE CUBOIDAL EPITHELIUM

Examples Throughout the Body

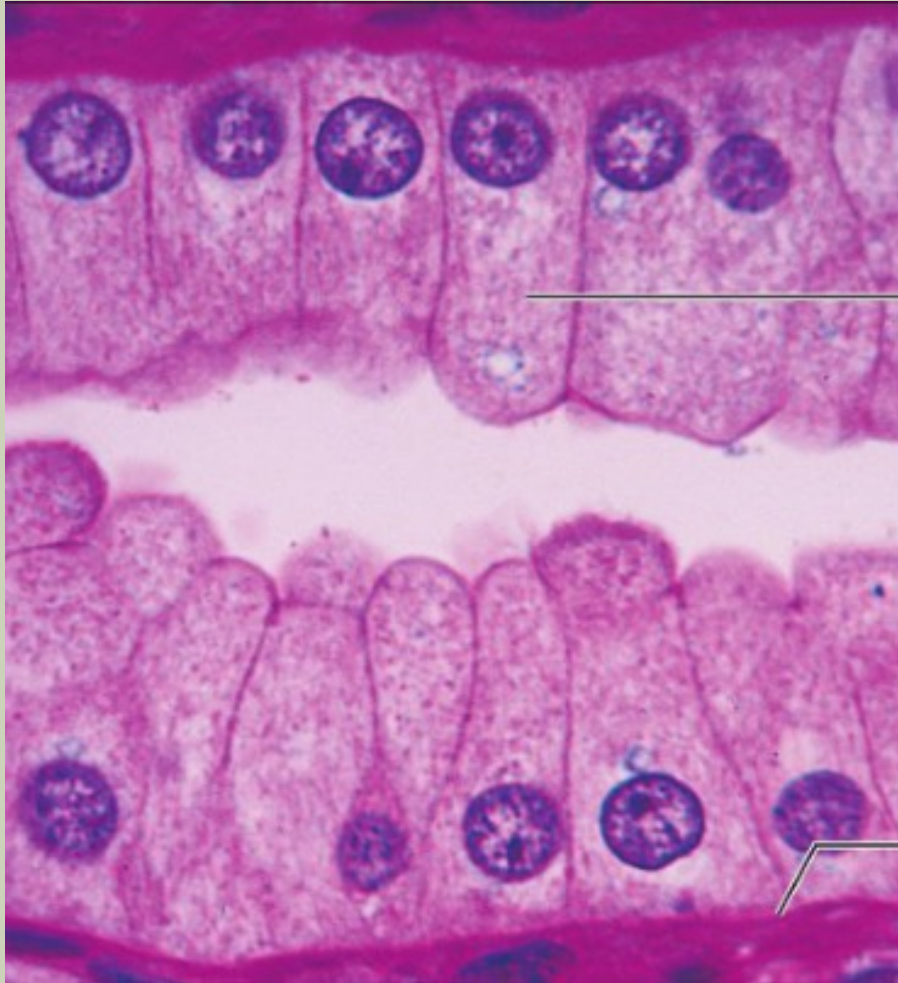
- Glandular epithelium
- Lining of glandular ducts.

Examples in the Eye

- Retinal pigment epithelium
- Pigmented epithelium of the ciliary body
- Non-pigmented epithelium of the ciliary body
- Pigmented epithelium of the iris

They tend to occur where there is rapid exchange of fluid / ions.

SIMPLE COLUMNAR EPITHELIUM



- Single layer of polarized cells resting on a basement membrane
- Cells are taller than they are wide
- In cross-section, the nuclei are round or oval and placed near the basal surface.

SIMPLE COLUMNAR EPITHELIUM

Examples Throughout the Body

- Lining of Uterus
- Lining of GI system

Examples in the Eye

- None

They tend to occur where there exchange of fluid or metabolites

CILIATED SIMPLE COLUMNAR EPITHELIUM



- Rare
- Prime example is the uterine (Fallopian) tube

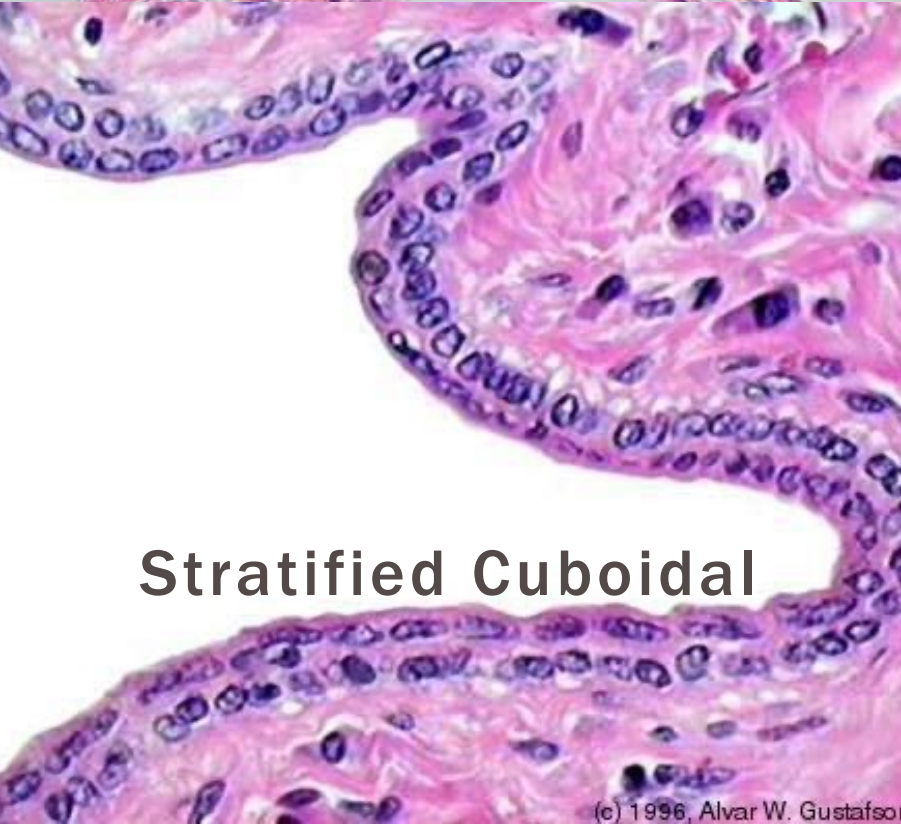
NOMENCLATURE OF STRATIFIED EPITHELIA



Stratified Squamous

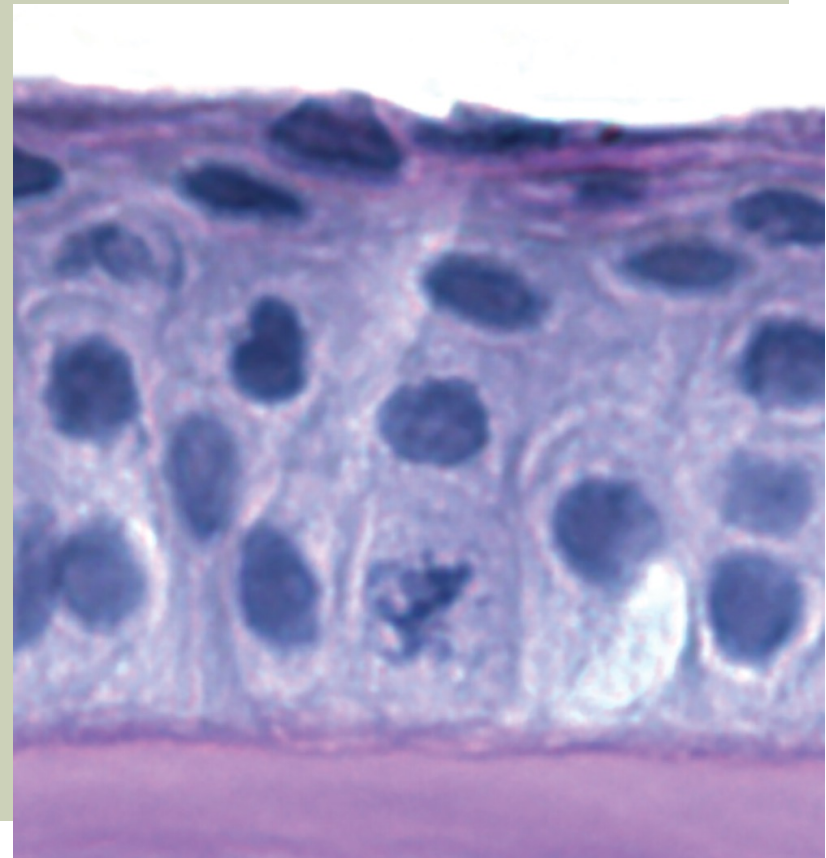
A histological section of stratified squamous epithelium, likely from the skin. It shows multiple layers of cells. The surface layer consists of flattened, squamous cells. Deeper layers contain more rounded cells. There are some keratin-filled cysts visible.

Named for the morphology of the surface cells. If normal, mitotic division found **ONLY** in the basal layer



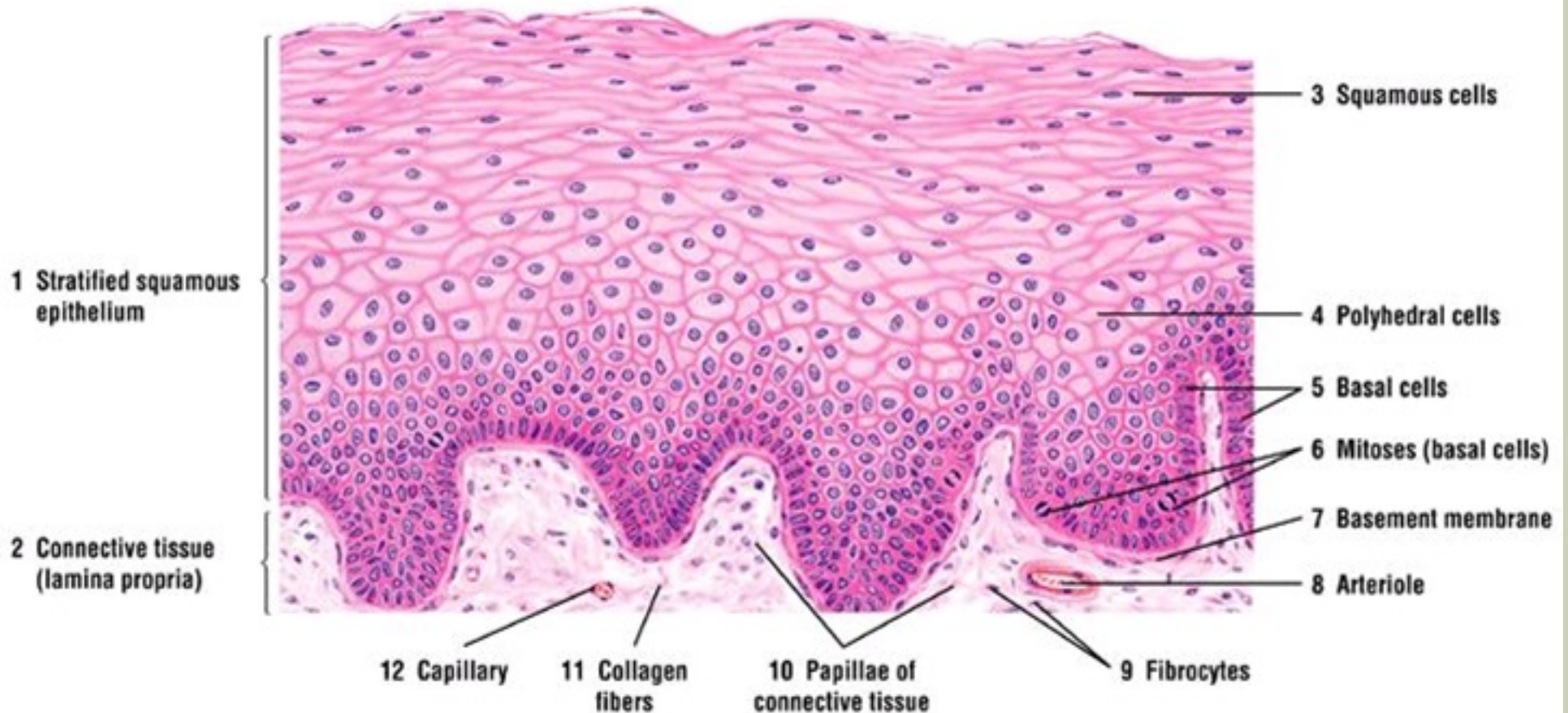
Stratified Cuboidal

A histological section of stratified cuboidal epithelium. It shows two distinct layers of cells. The surface layer is composed of a single layer of cuboidal cells. The underlying layer is also composed of cuboidal cells, but they are arranged in a more disorganized manner.



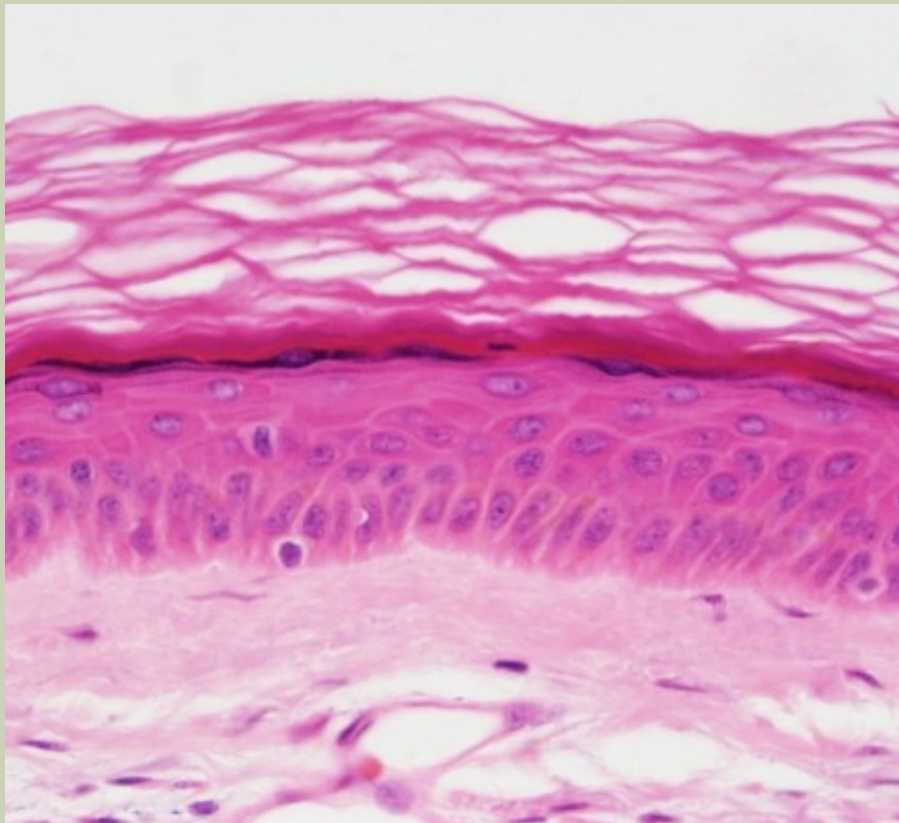
STRATIFIED SQUAMOUS EPITHELIUM

- This type of epithelium is well suited to areas in the body subject to constant abrasion, as it is the thickest and layers can be sequentially sloughed off and replaced before the basement

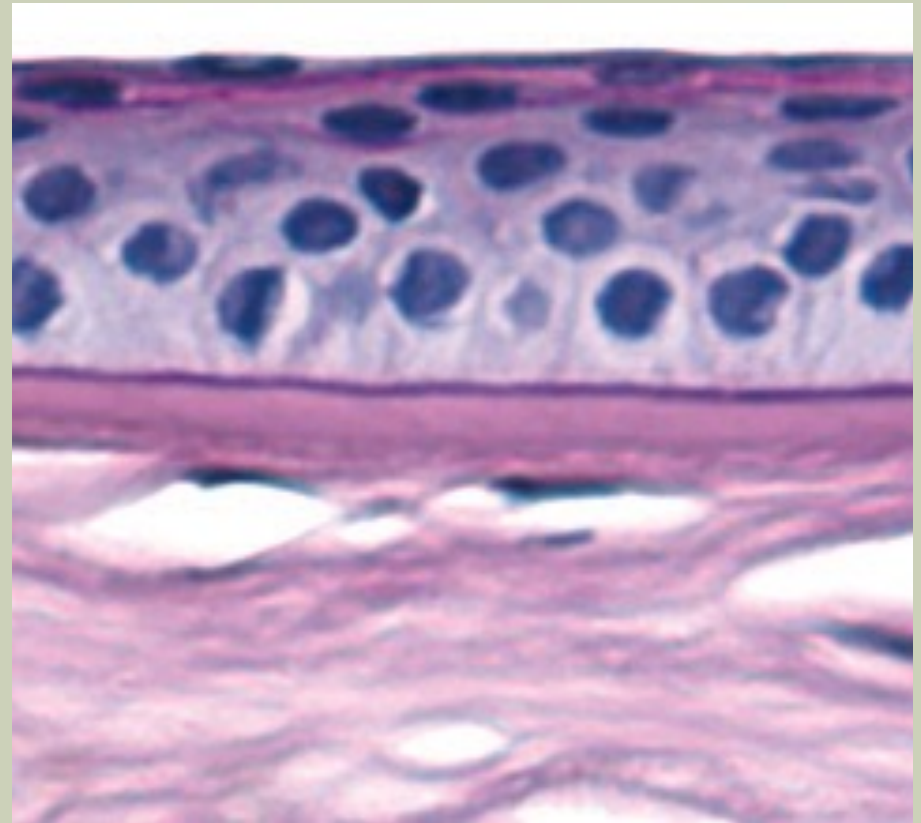


KERATINIZED VS NON-KERATINIZED STRATIFIED SQUAMOUS EPITHELIUM

Keratinized - skin

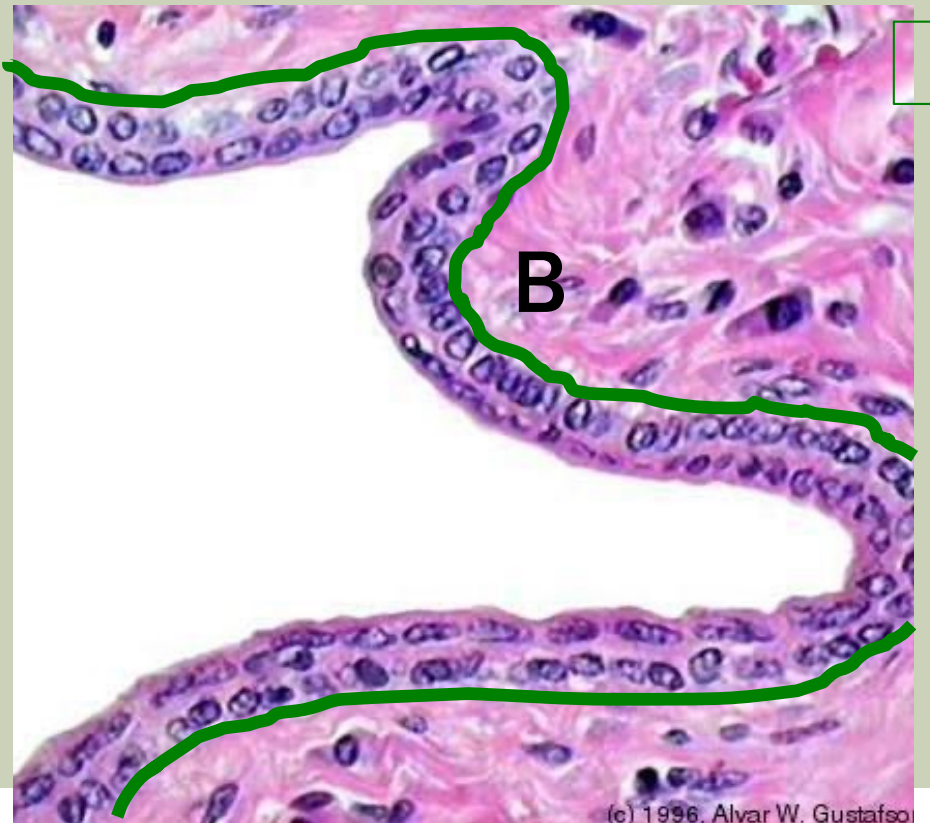


**Non-Keratinized - surface of
cornea, inner lining of the mouth,
esophagus, anal canal and vagina**

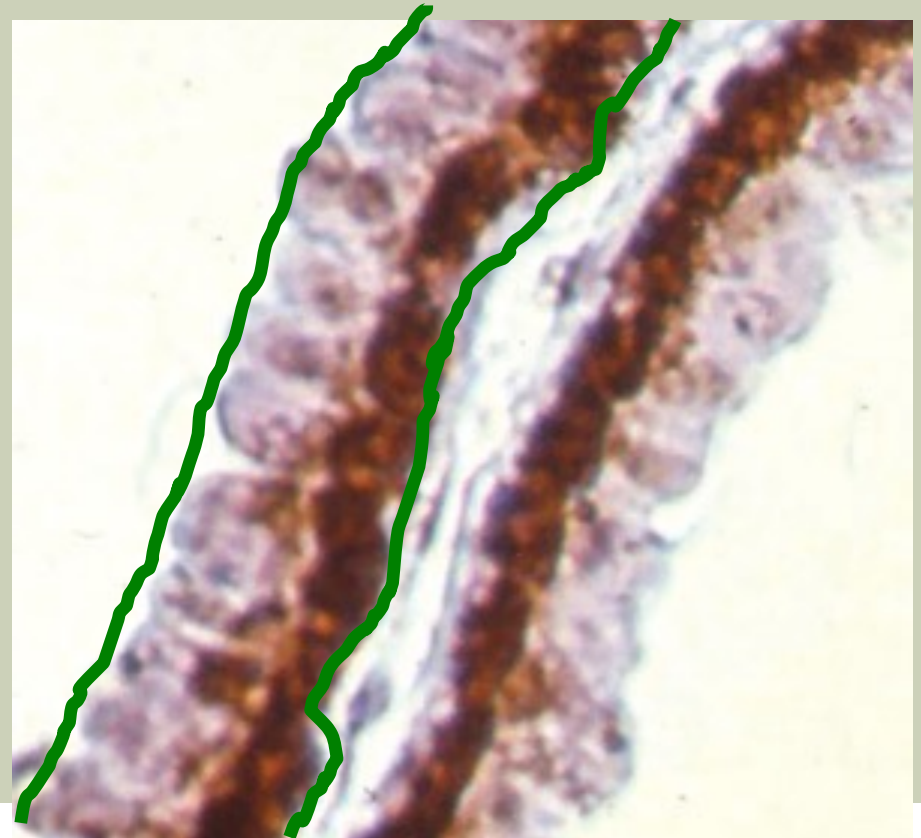


STRATIFIED CUBOIDAL

True stratified Cuboidal



Bi-layers that look stratified but are not

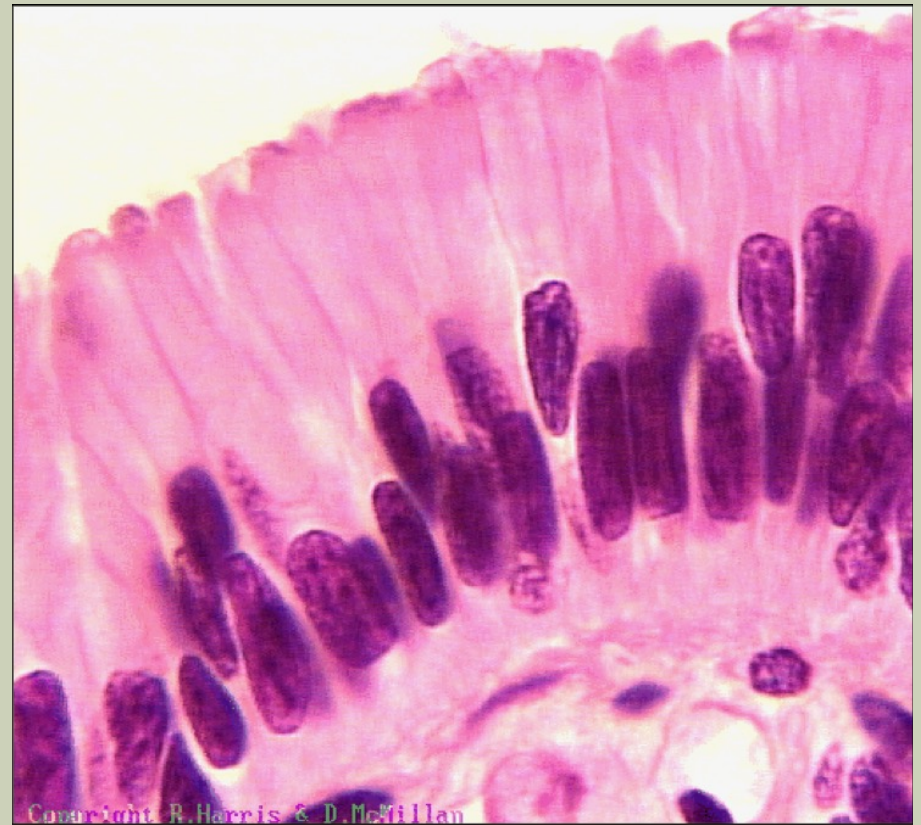


PSEUDOSTRATIFIED COLUMNAR EPITHELIUM

Ciliated – respiratory system

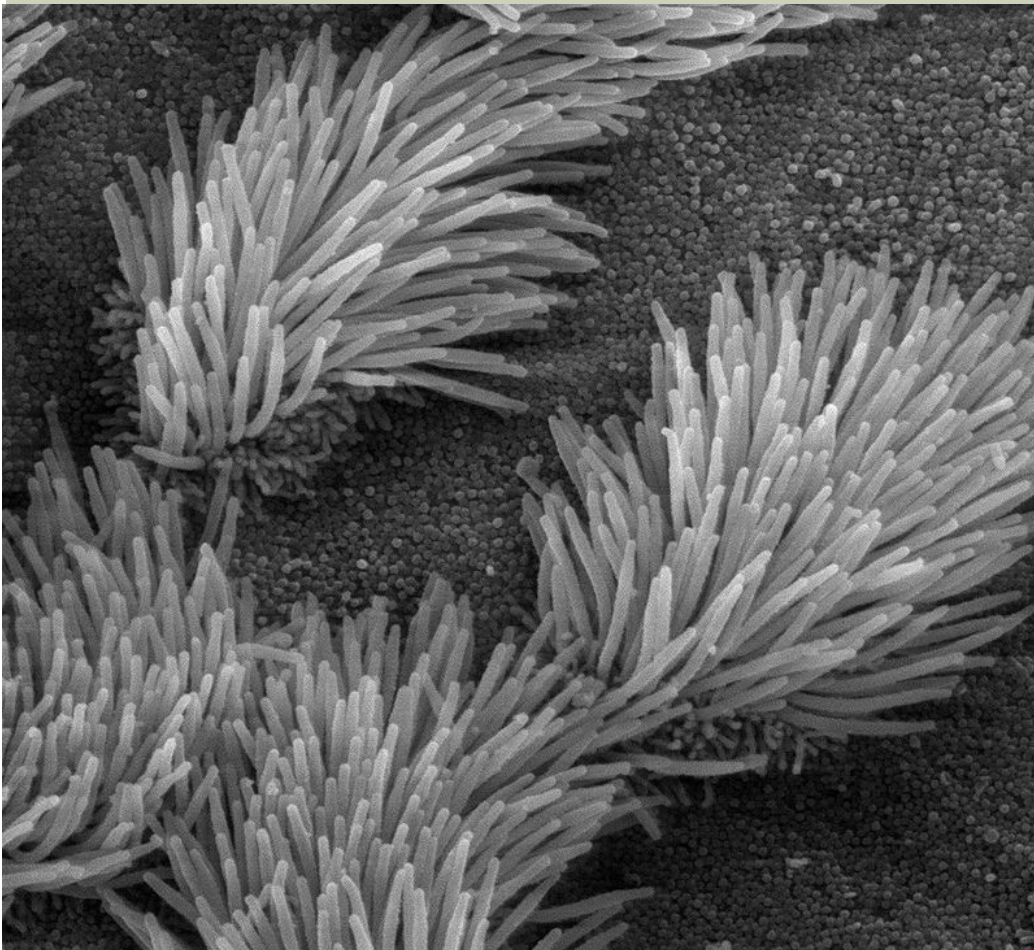


Non-ciliated

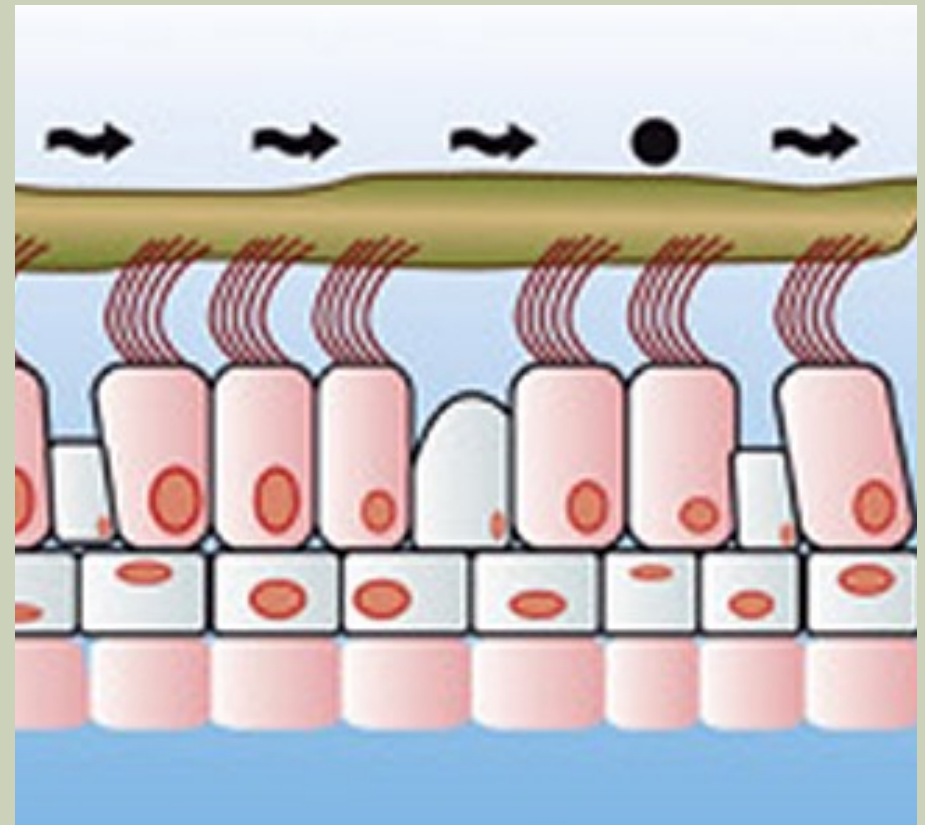


CILIA

Tufts of Tracheal Cilia

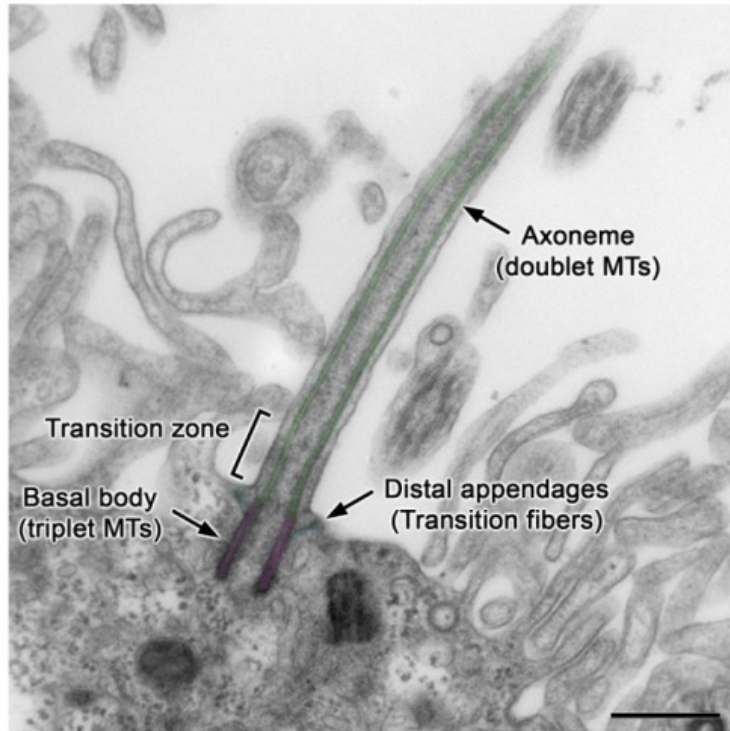


Cilia move mucus



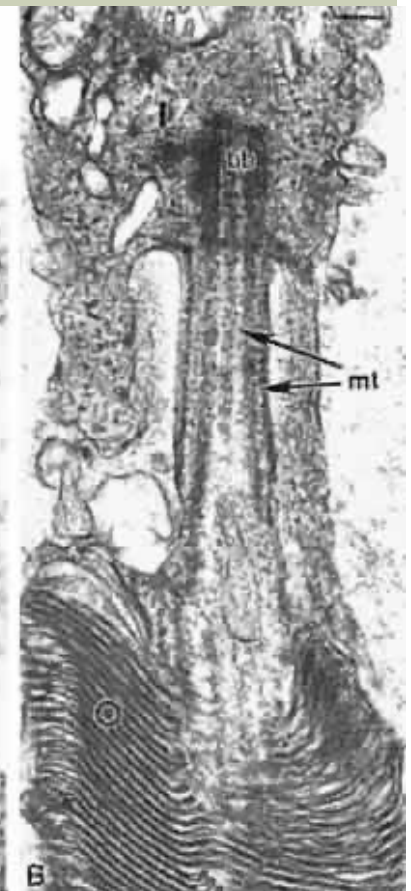
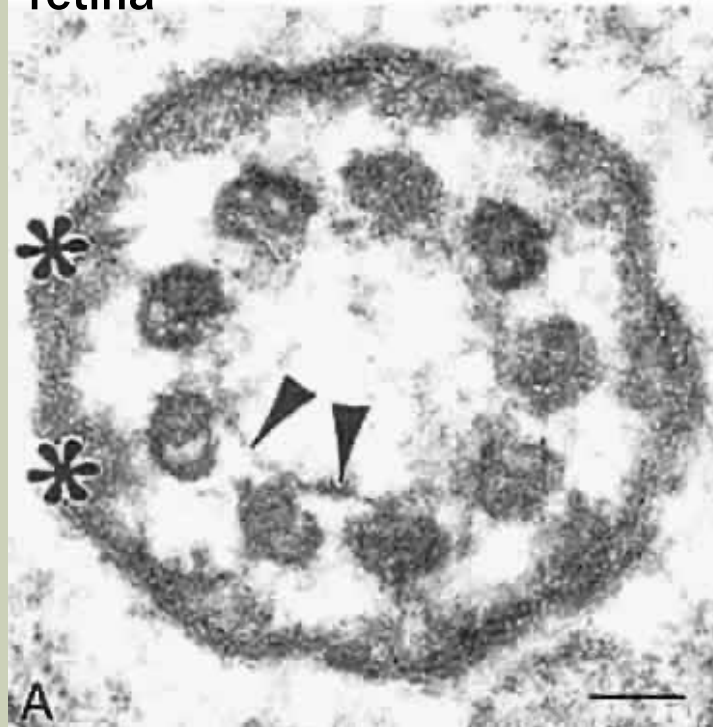
CILIA – MOTILE CILIA MOVE MUCUS AND ALSO MOVE SPERM CELLS

A

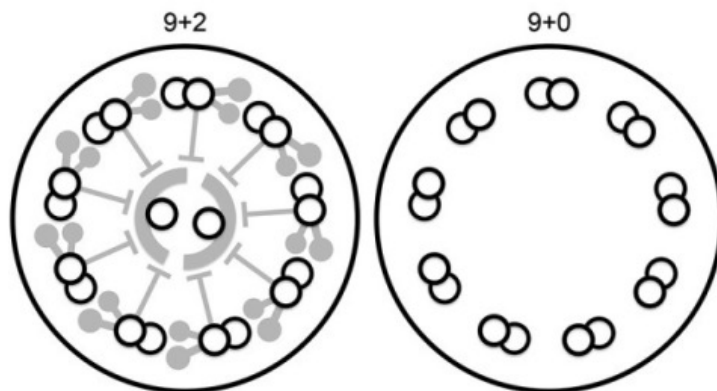


- Motile cilia have 9+2 microtubular doublets
- Non-motile (sensory) cilia are 9+0

Non-motile cilia are found in photoreceptor cells of the retina



B



BARDET-BIEDL SYNDROME – ONE OF SEVERAL INHERITED CILIOPATHIES

Extra digit adjoining little finger or baby toe



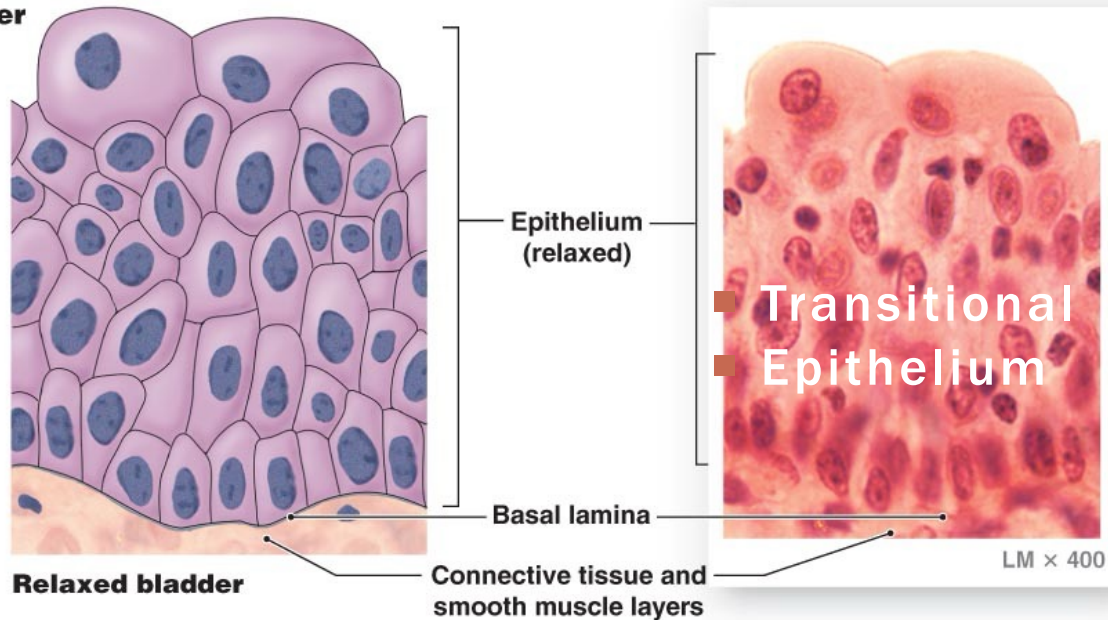
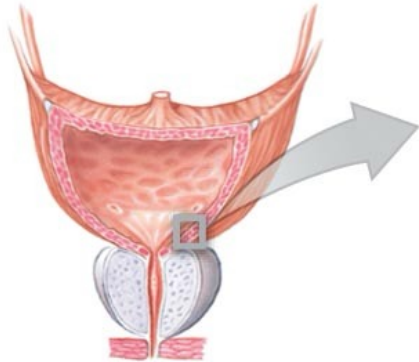
A variant of retinitis pigmentosa



The transitional epithelium in an empty and a full urinary bladder

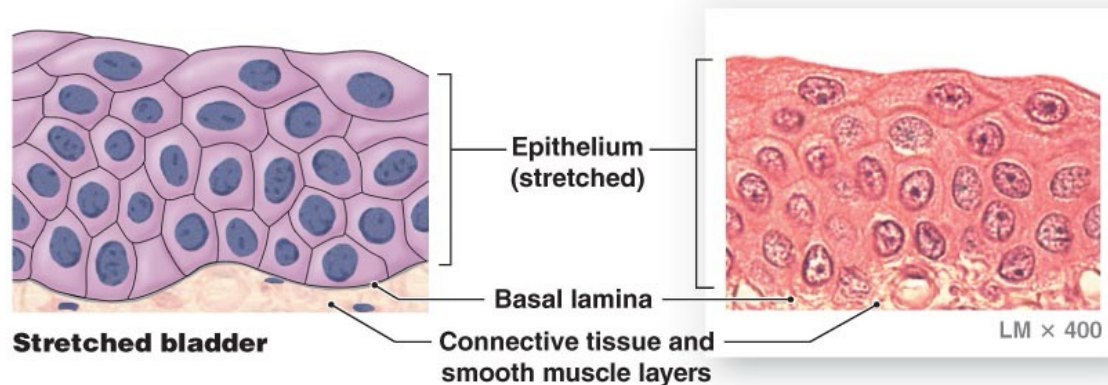
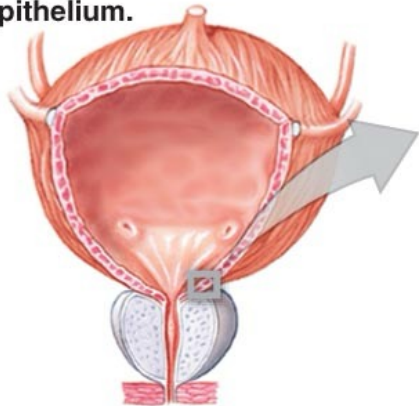
Epithelium in a Relaxed Bladder

In an empty urinary bladder, the superficial cells are cuboidal with a dome-shaped surface.



Epithelium in a Stretched Bladder

When the urinary bladder is full, the volume of urine has stretched the lining to such a degree that the epithelium appears flattened, and more like a stratified squamous epithelium.



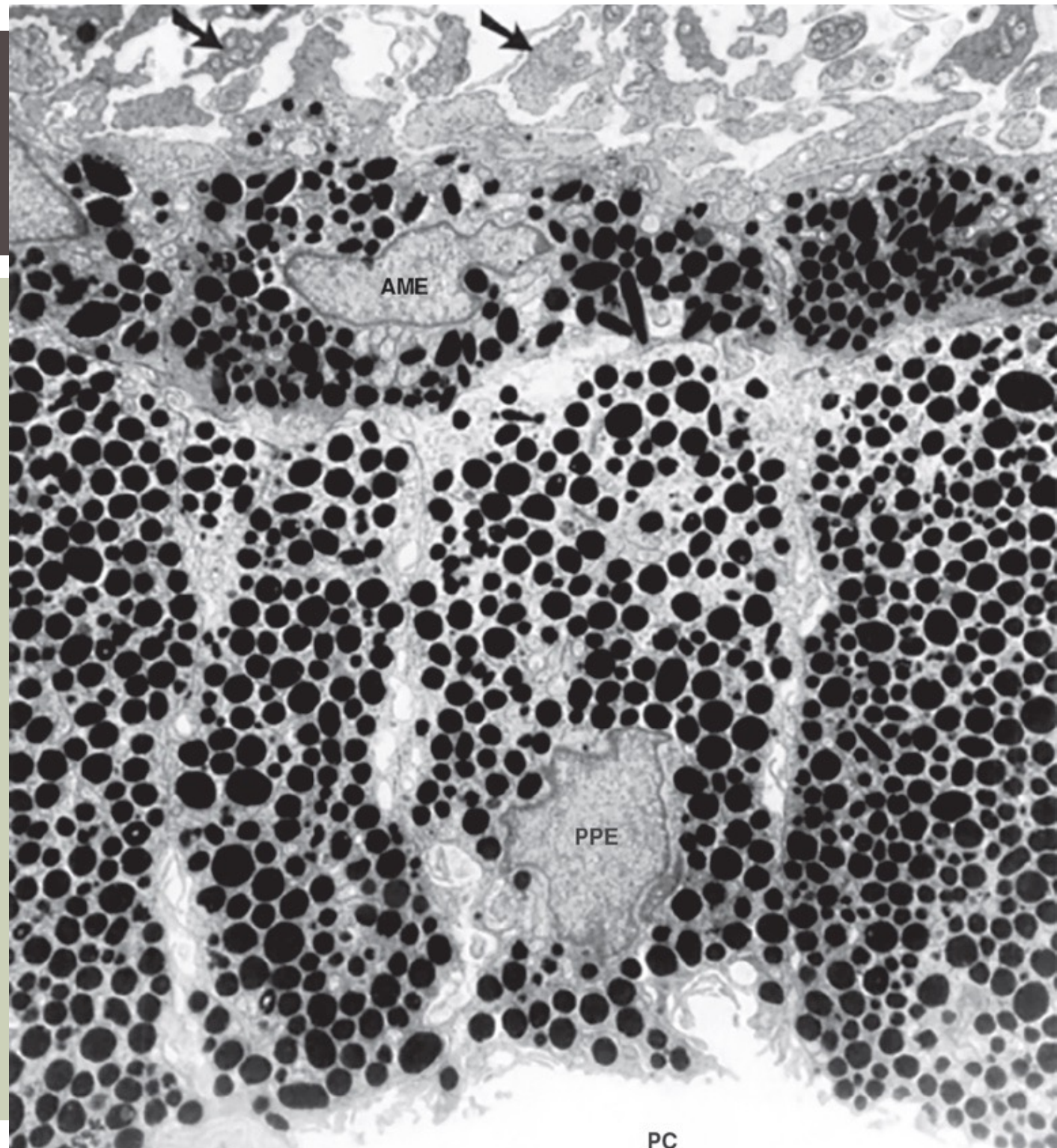
Transitional Epithelium

- Found in: renal pelvis, urinary bladder, the ureters, the superior urethra, and the prostatic and ejaculatory ducts of the prostate.

TRANSITIONAL EPITHELIUM -
SOMETIMES CALLED UROEPITHELIAL
CELLS

MYOEPIHELUM

- They are found in various exocrine glands, especially around ducts
sweat
mammary
lacrimal
salivary
- A layer of pigmented myoepithelial cells also forms the dilator pupillae muscle in the iris.

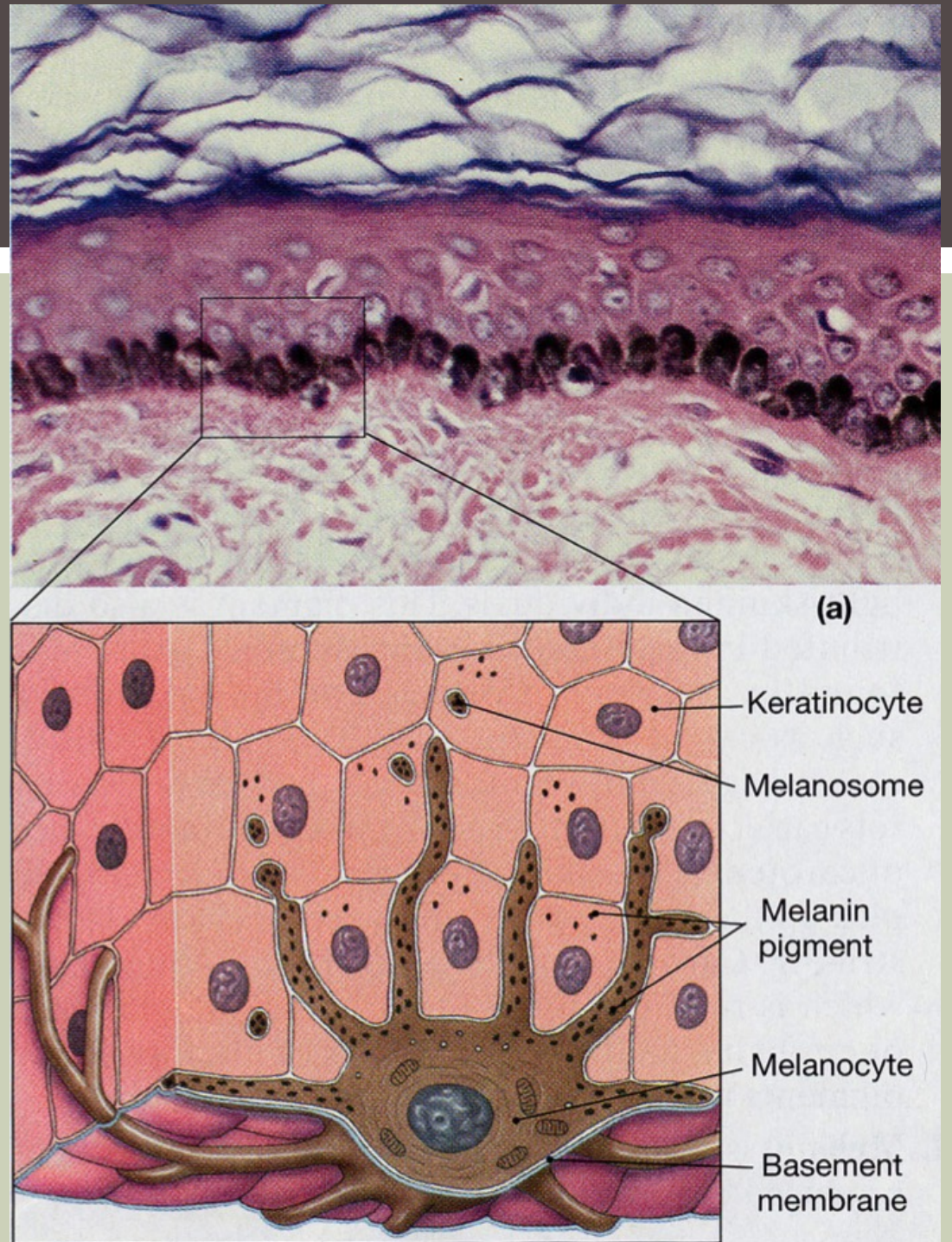


CELLS THAT INTERSPERSE WITHIN EPITHELIA

- Epithelia are rarely composed purely of the epithelial cells themselves.
- Melanocytes among basal cells– Skin and Conj -pigmentation
- Langerhan's cells – Skin and Conj - Antigen recognition and processing
- Goblet cells – Conj, GI and respiratory epithelia - unicellular mucus secreting glands
- M cells – various mucosa – antigen recognition and presentation
- Many others.....

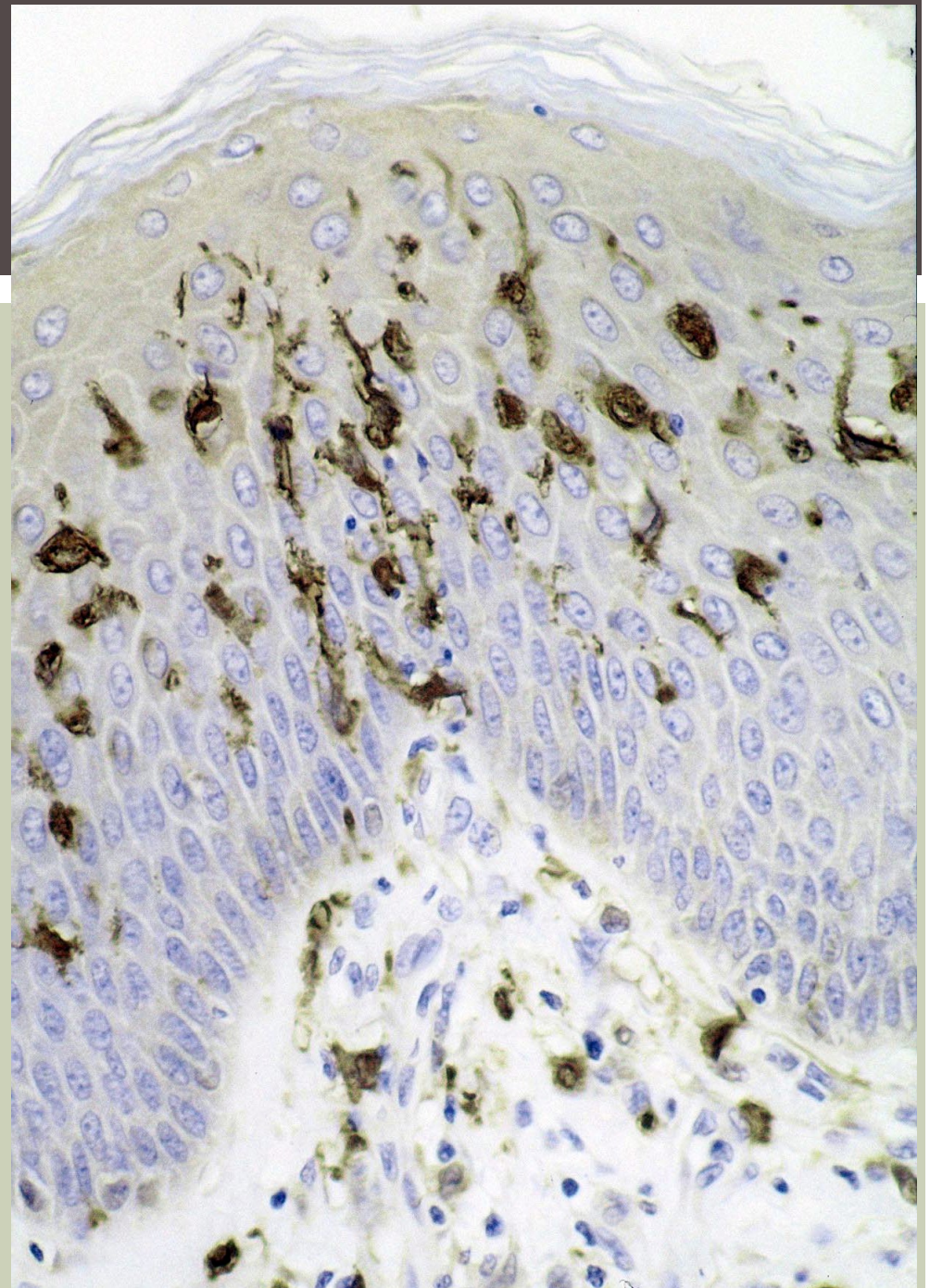
MELANOCYTES

Keratinocytes do not make melanin.
Melanocytes distributed within the basal layers extend numerous processes that infuse formed pigment into the differentiating epithelial cells

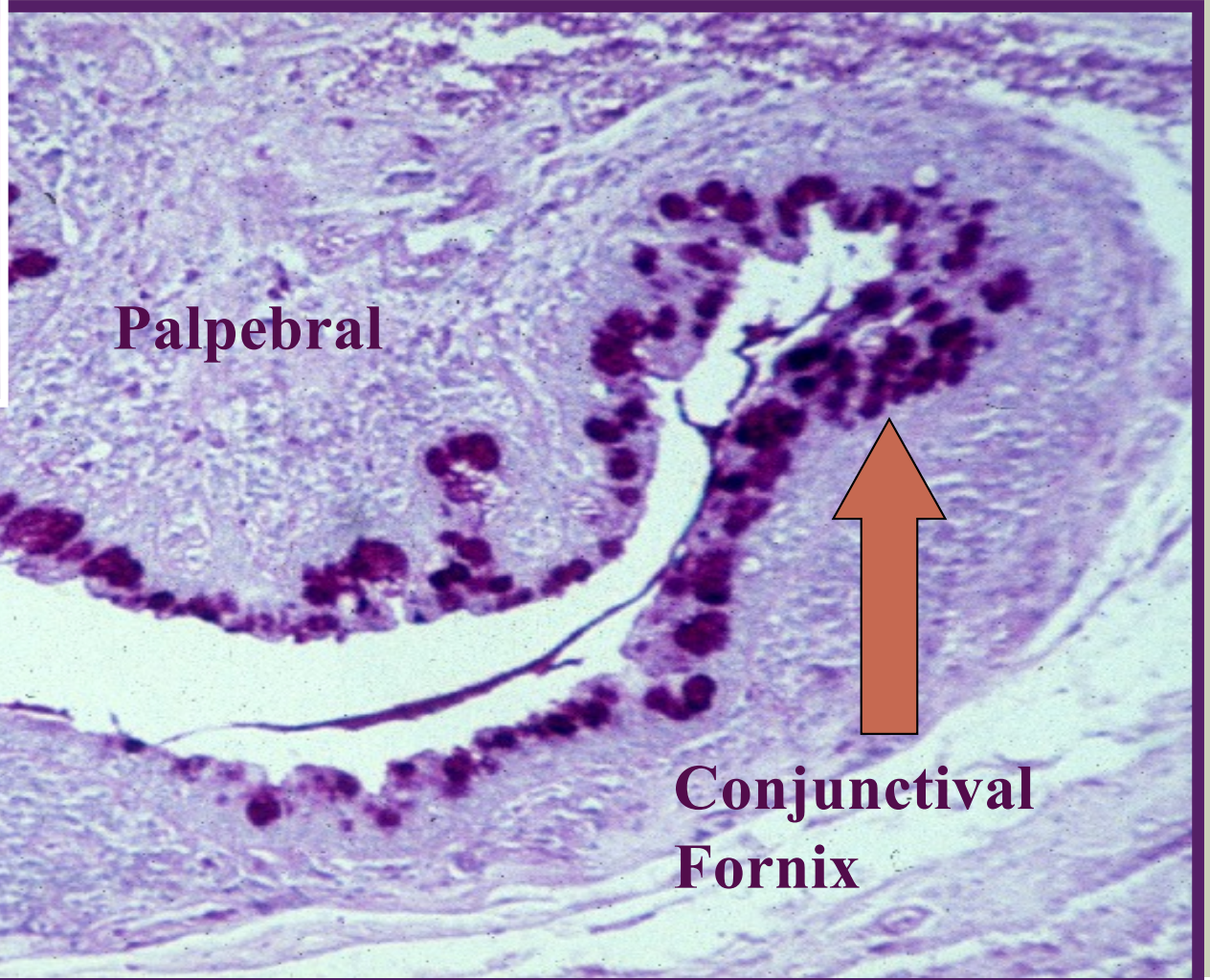


LANGERHANS CELLS

Langerhans cells are found in certain but not all epithelia, where they play a central role in antigen recognition, processing and presentation to the subepithelial immune system. Very common in skin and conj. Some in peripheral cornea. NONE in central cornea to ensure that severe inflammation that would cause scarring is mitigated.



GOBLET CELLS



Palpebral

Bulbar

**Conjunctival
Fornix**

INTERCELLULAR JUNCTIONS

- Zonulae Occludentes
 - (tight junctions)

- Zonulae adherentes

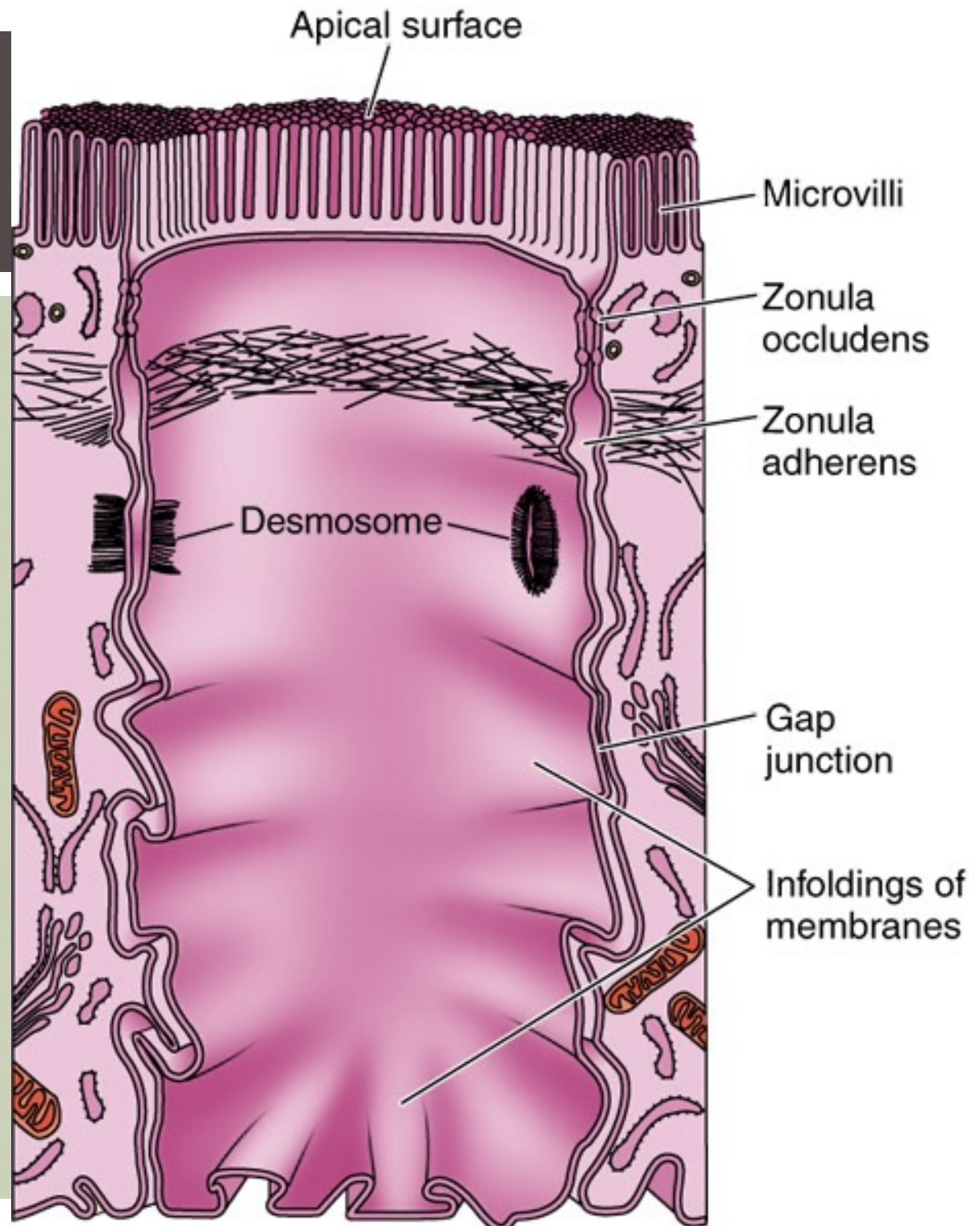
Note that zonular junctions form a continuous belt around the entire circumference of each cell.

- Gap junctions

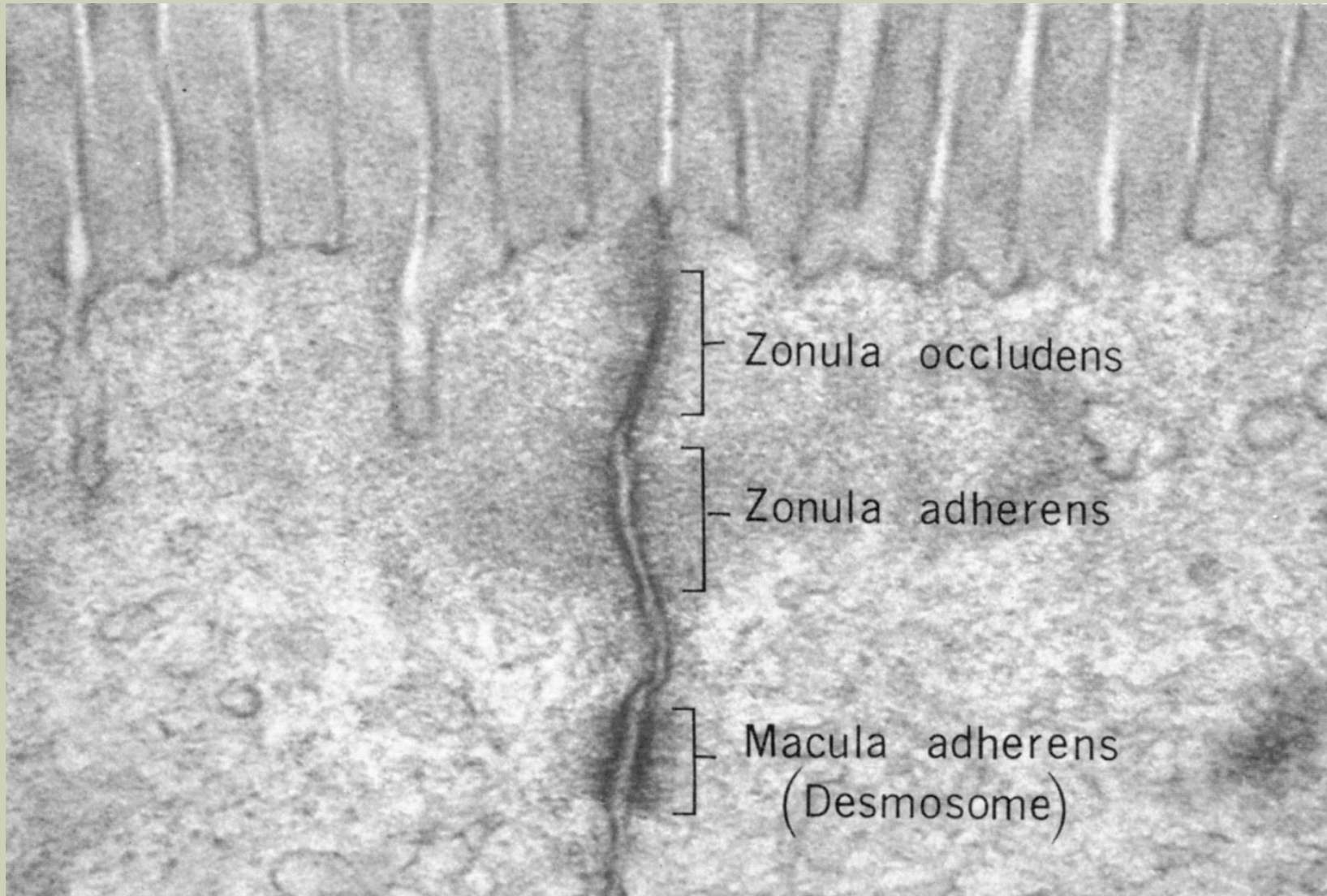
- Desmosomes

- Hemi-desmosomes

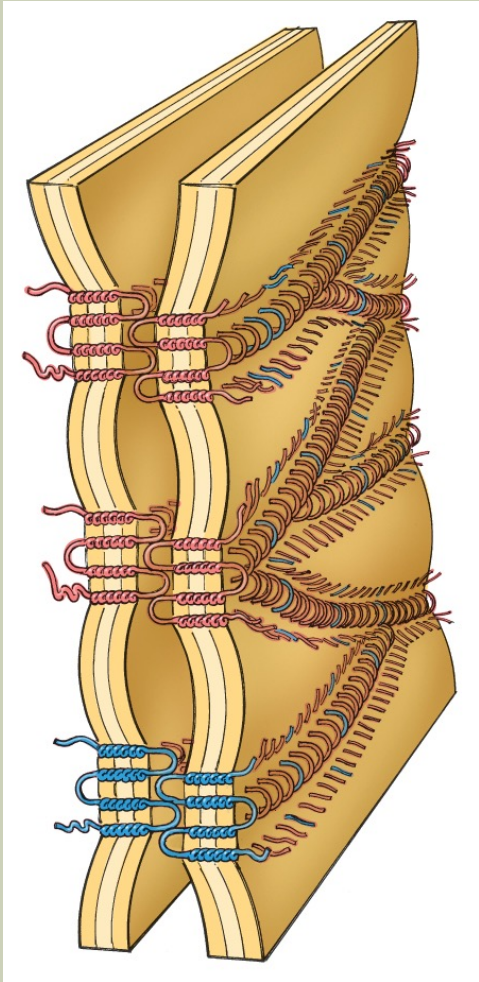
- The above three are focal junctions



IN EPITHELIA, THESE JUNCTIONS OFTEN APPEAR TOGETHER

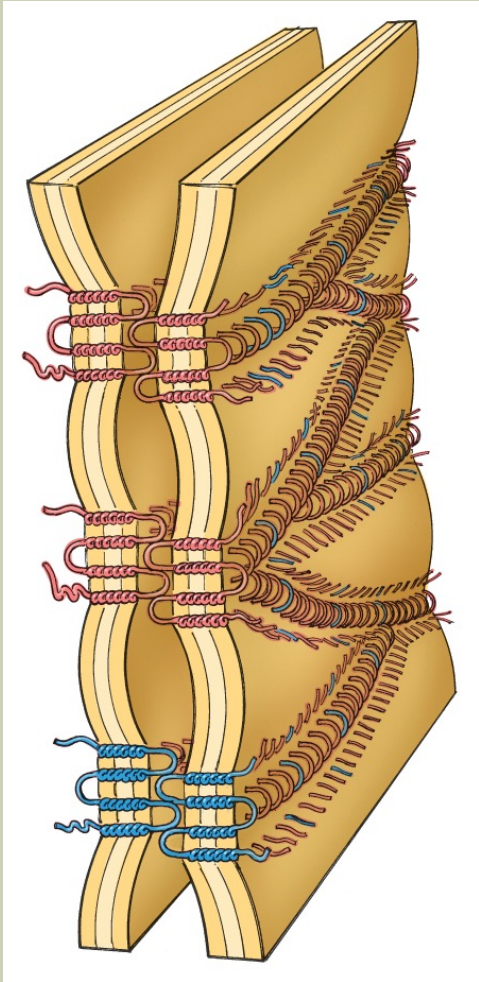


ZONULA OCCLUDENS – PLURAL IS ZONULAE OCCLUDENTES (TIGHT JUNCTIONS)

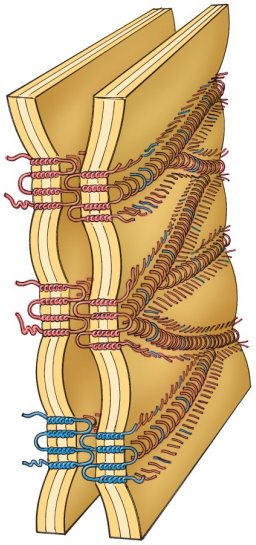


- Tight junctions are belt-like junctions and ring the entire circumference of the cell, sealing the intercellular cleft between the cell and all of its adjoining neighbors.
- Found only in epithelial/endothelial tissues.

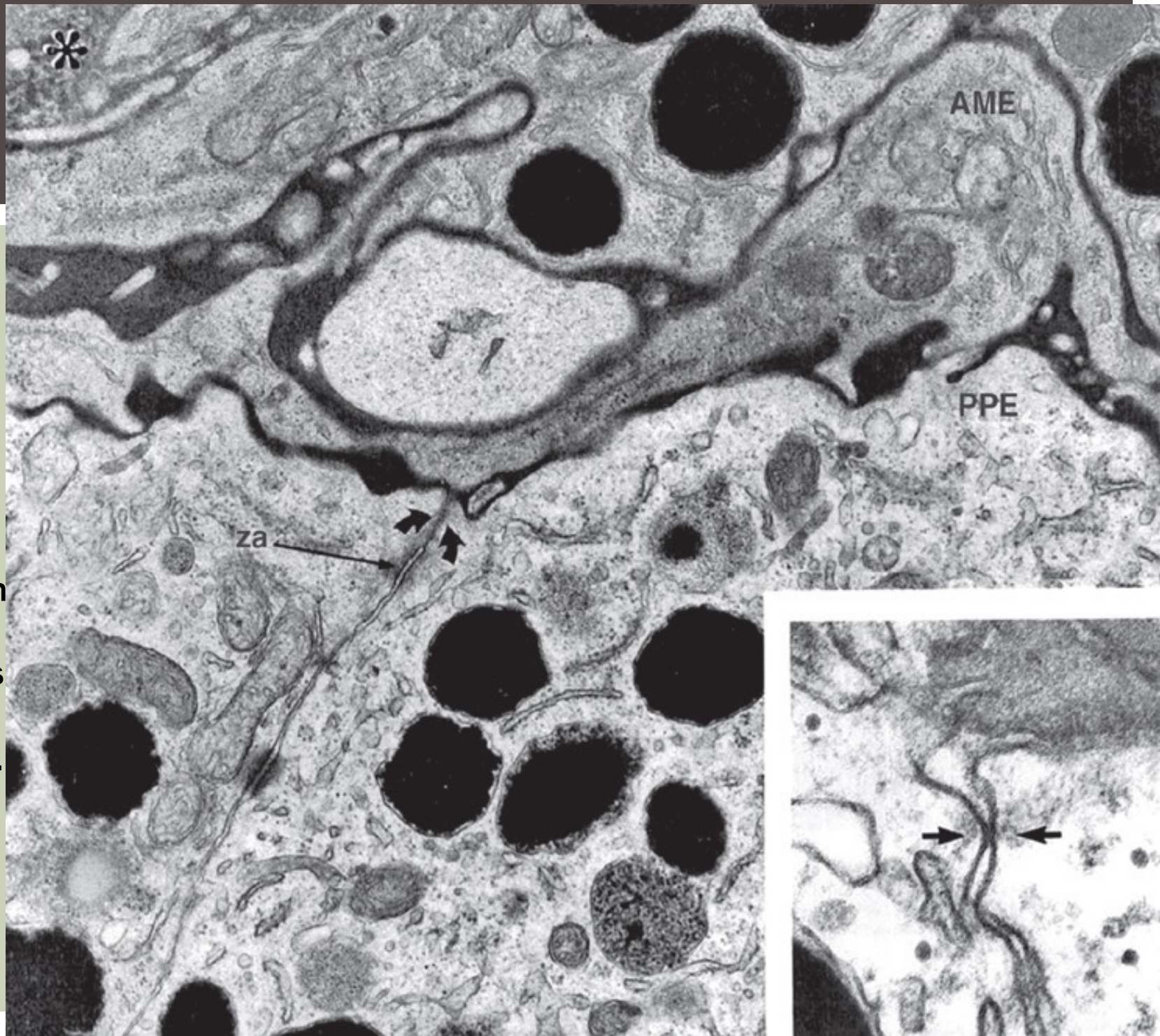
ZONULA OCCLUDENS – PLURAL IS ZONULAE OCCLUDENTES (TIGHT JUNCTIONS)



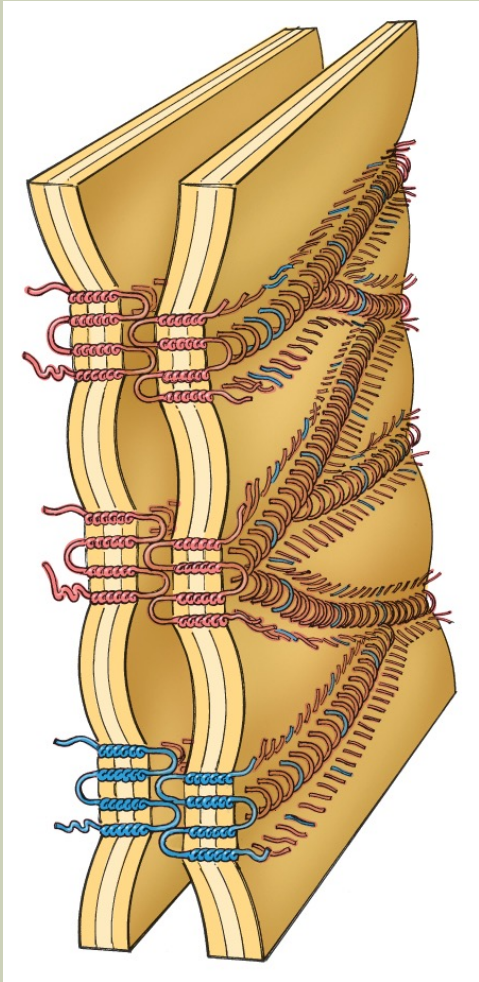
- These junctions provide a barrier function, serving purposes such as material transport and maintenance of osmotic balance.
- Tight Junctions prevent the passage of molecules and ions through the space between plasma membranes of adjacent cells, so materials must actually enter the cells (by diffusion or active transport).
- Tight Junctions help to maintain the polarity of cells by preventing the lateral diffusion of integral membrane proteins between the apical and lateral/basal surfaces, allowing the specialized functions of each surface (for example receptor-mediated endocytosis at the apical surface and exocytosis at the basolateral surface) to be preserved.
- This aims to preserve the transcellular transport.



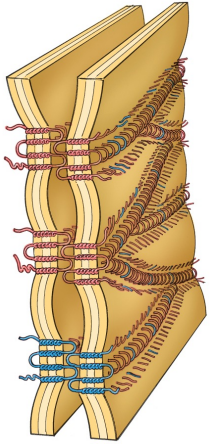
Demonstration of ability of tight junction to prevent non-specific diffusion of macromolecules such as plasma-derived proteins.



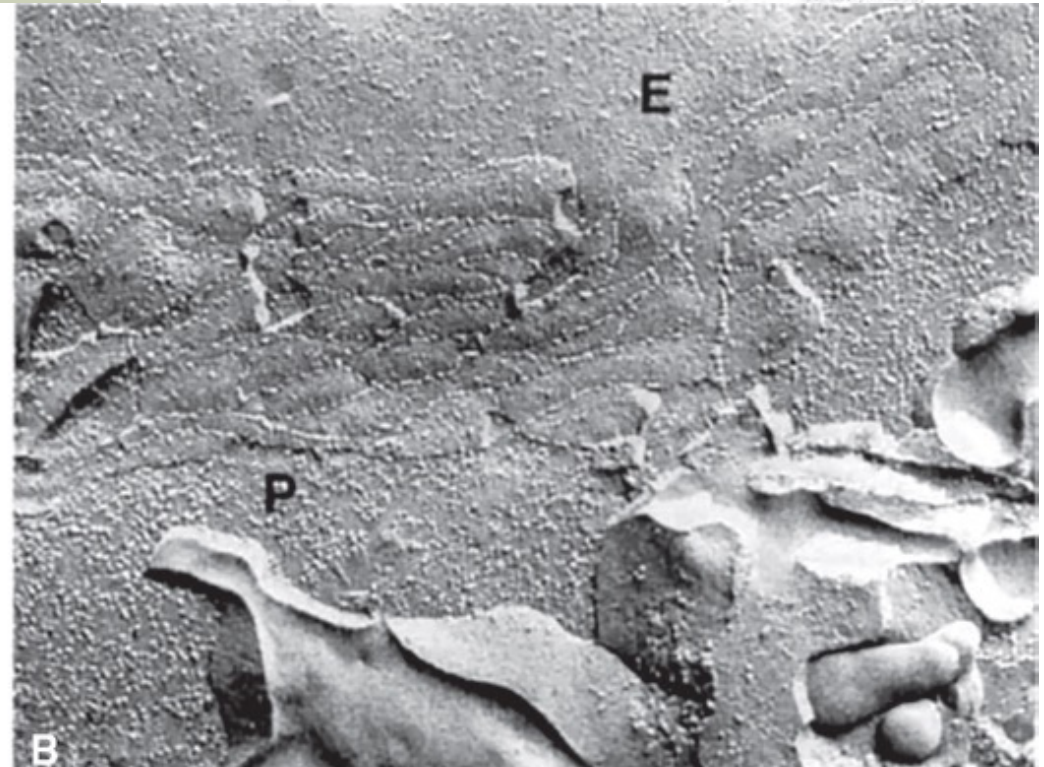
ZONULA OCCLUDENS – PLURAL IS ZONULAE OCCLUDENTES (TIGHT JUNCTIONS)

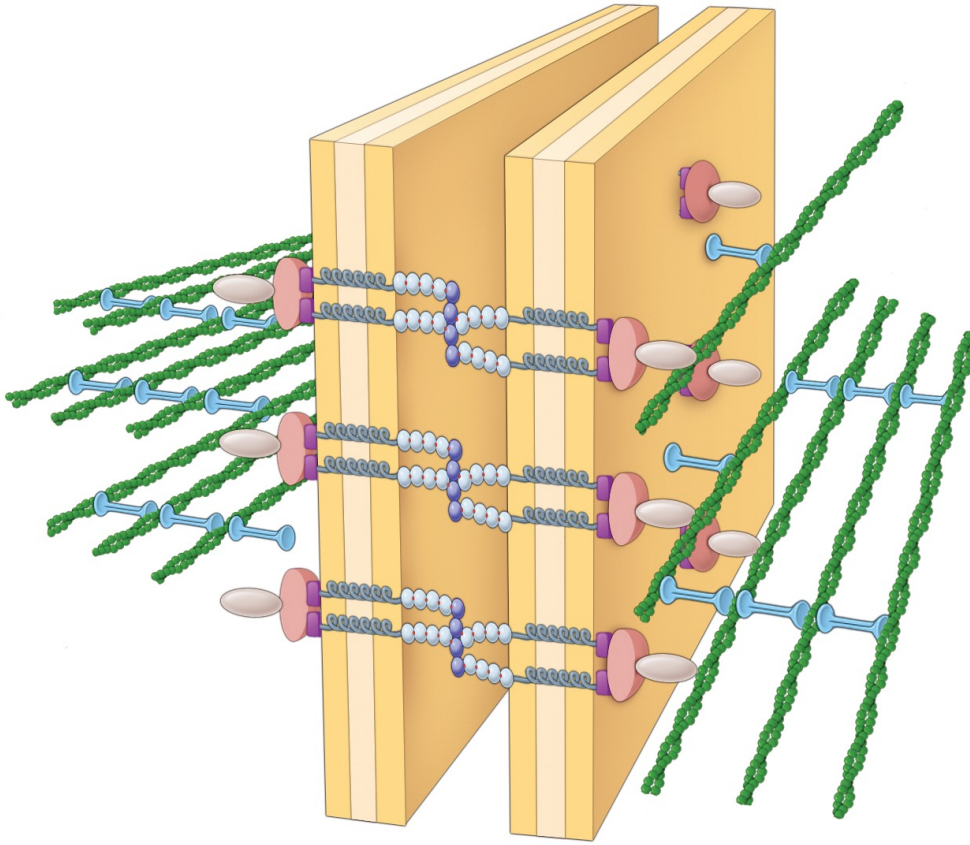


- Formed by specialized string of intramembranous proteins that fuse with a similar pattern of proteins in the membranes of adjoining cells.
- The occluding protein strings are formed of proteins from one of two protein families, including the claudins and the occludins. These intramembranous proteins associate with different peripheral membrane proteins such as ZO-1 located on the intracellular side of plasma membrane.
- The peripheral proteins anchor strands to the actin component of the cytoskeleton. Thus, tight junctions join together the cytoskeletons of adjacent cells.



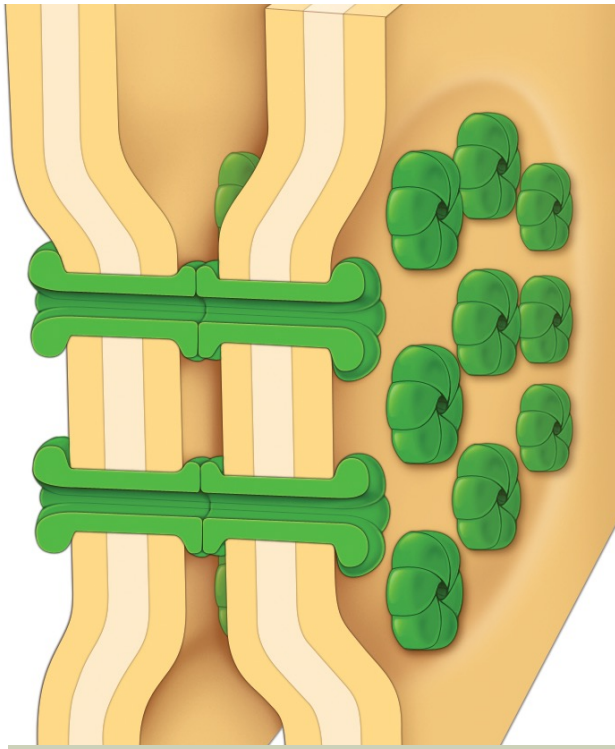
**ZONULA OCCLUDENS –
PLURAL IS ZONULAE
OCCLUDENTES
(TIGHT JUNCTIONS)**





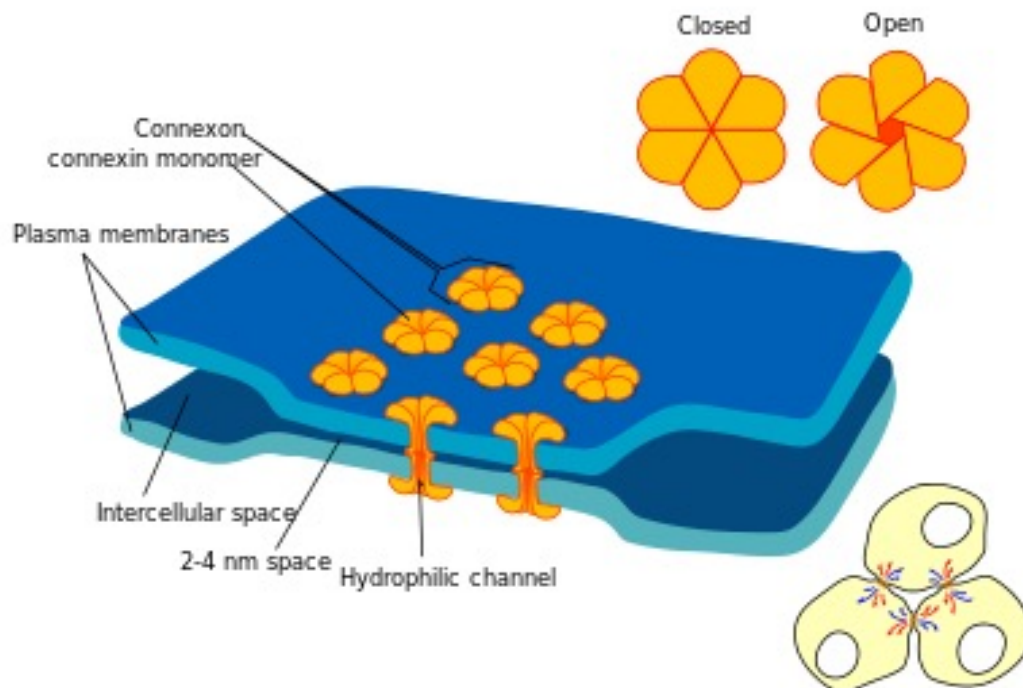
ZONULA ADHERENS – PLURAL IS ZONULAE ADHERENTES

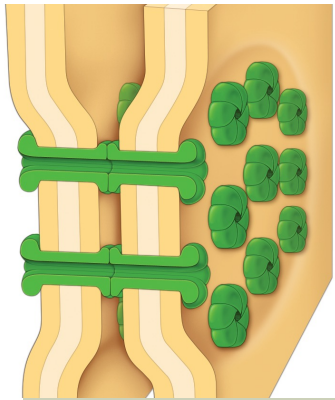
- Adherens junctions are composed of the following proteins:
- **Cadherins**. The cadherins are a family of transmembrane proteins that form homodimers in a calcium-dependent manner with other cadherin molecules on adjacent cells.
- **p120** (sometimes called **delta catenin**) binds the juxtamembrane region of the cadherin.
- **γ-catenin** or gamma-catenin (**plakoglobin**) binds the catenin-binding region of the cadherin.
- **α-catenin** or alpha-catenin binds the cadherin indirectly via **β-catenin** or **plakoglobin** and links the actin cytoskeleton with cadherin.
- They bind the F-actin cytoskeleton of adjacent cells to maintain tension and adhesion between sheets of cells.



GAP JUNCTIONS

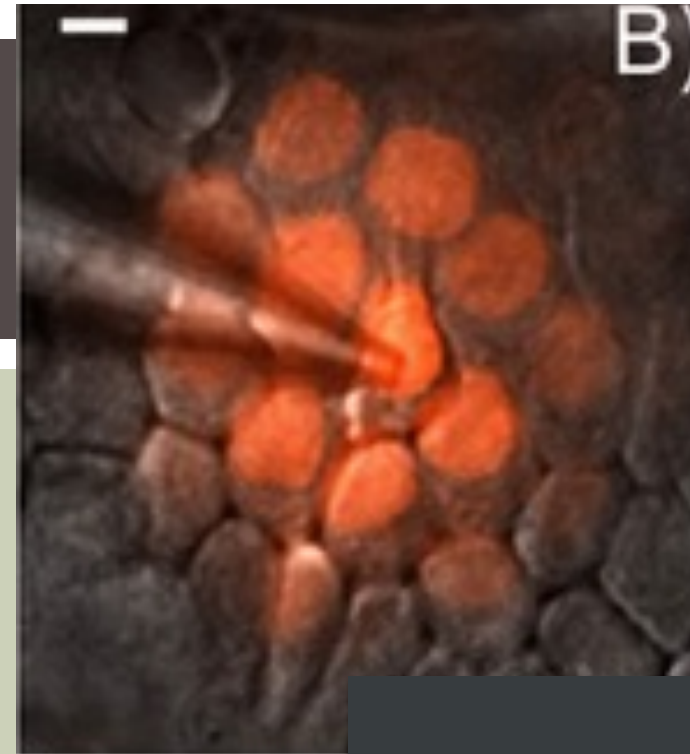
- Focal junctions provide intercellular gated channels
- They facilitate electrotonic and metabolic coupling of cells to coordinate functions.
- Six proteins in the membrane form a ring channel
- These proteins are all one or another of a protein family known as connexins.

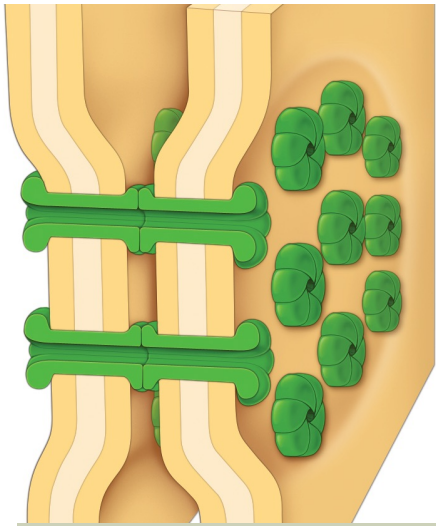




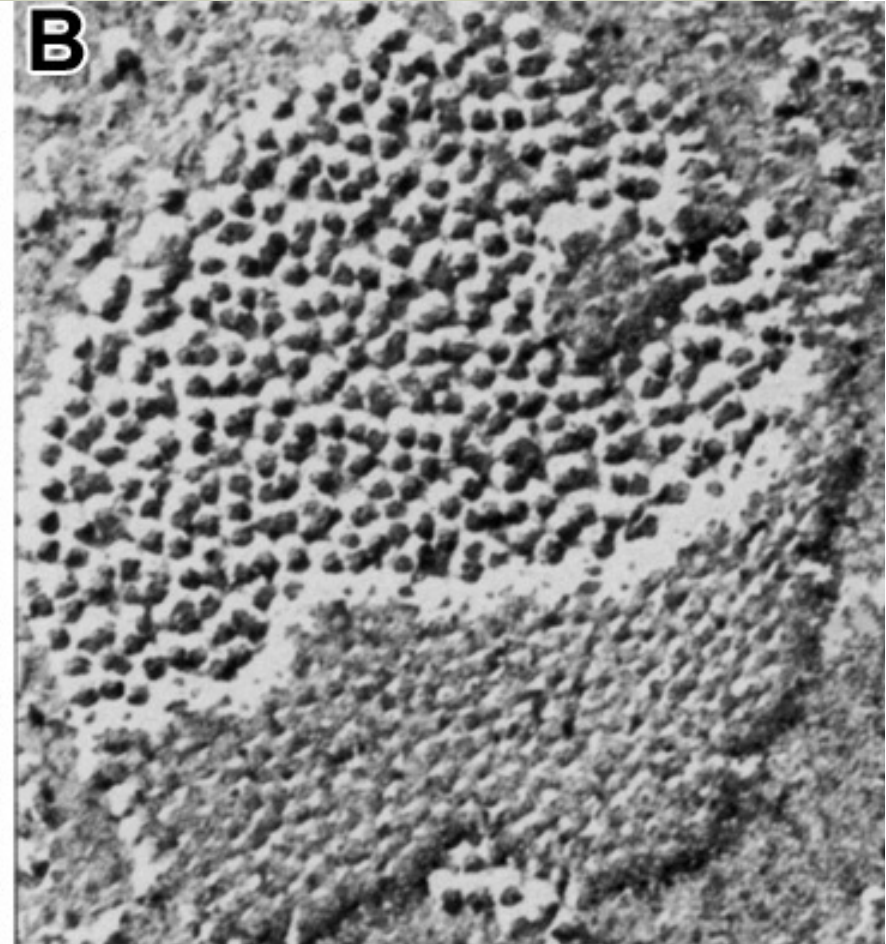
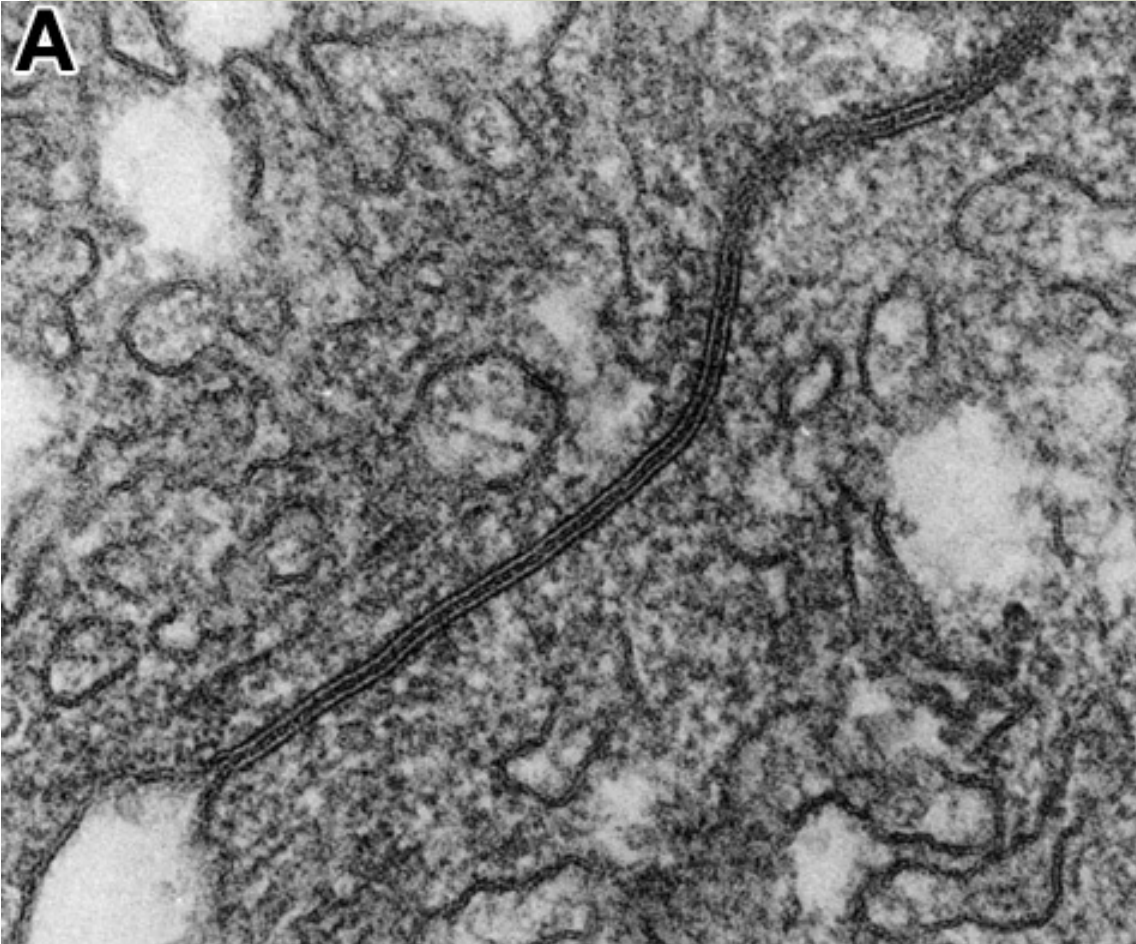
GAP JUNCTIONS

- They facilitate electrotonic and metabolic coupling of cells to coordinate functions.
- Note injection of dye into a single cell transfers to all surrounding cells via gap junctions (B)
- But if gap junctions are closed, no transfer occurs (D)

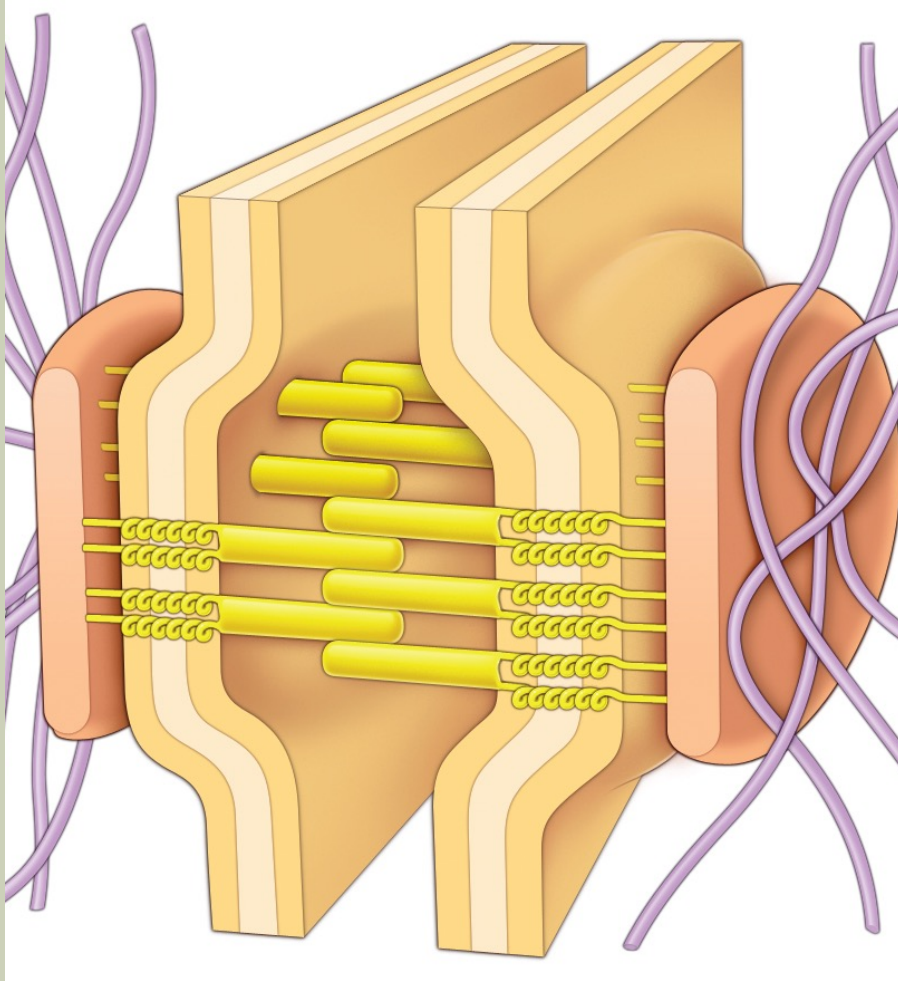




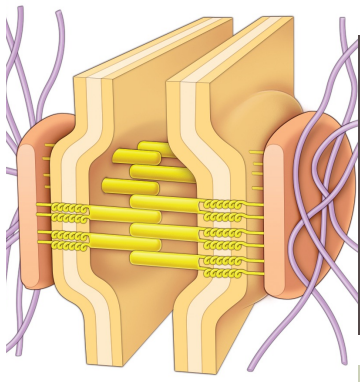
GAP JUNCTIONS



DESMOSOMES (MACULA ADHERENS)



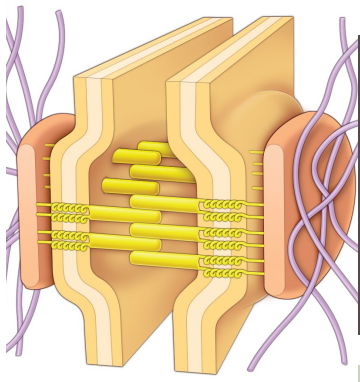
- Focal junctions serve as spot welds between cells and as anchor points for attachment of cytoskeleton.
- Desmosomes help to resist shearing forces and are found in simple and stratified squamous epithelium.
- Desmosomes are also found in muscle tissue where they bind muscle cells to one another.



DESMOSOMES (MACULA ADHERENS)

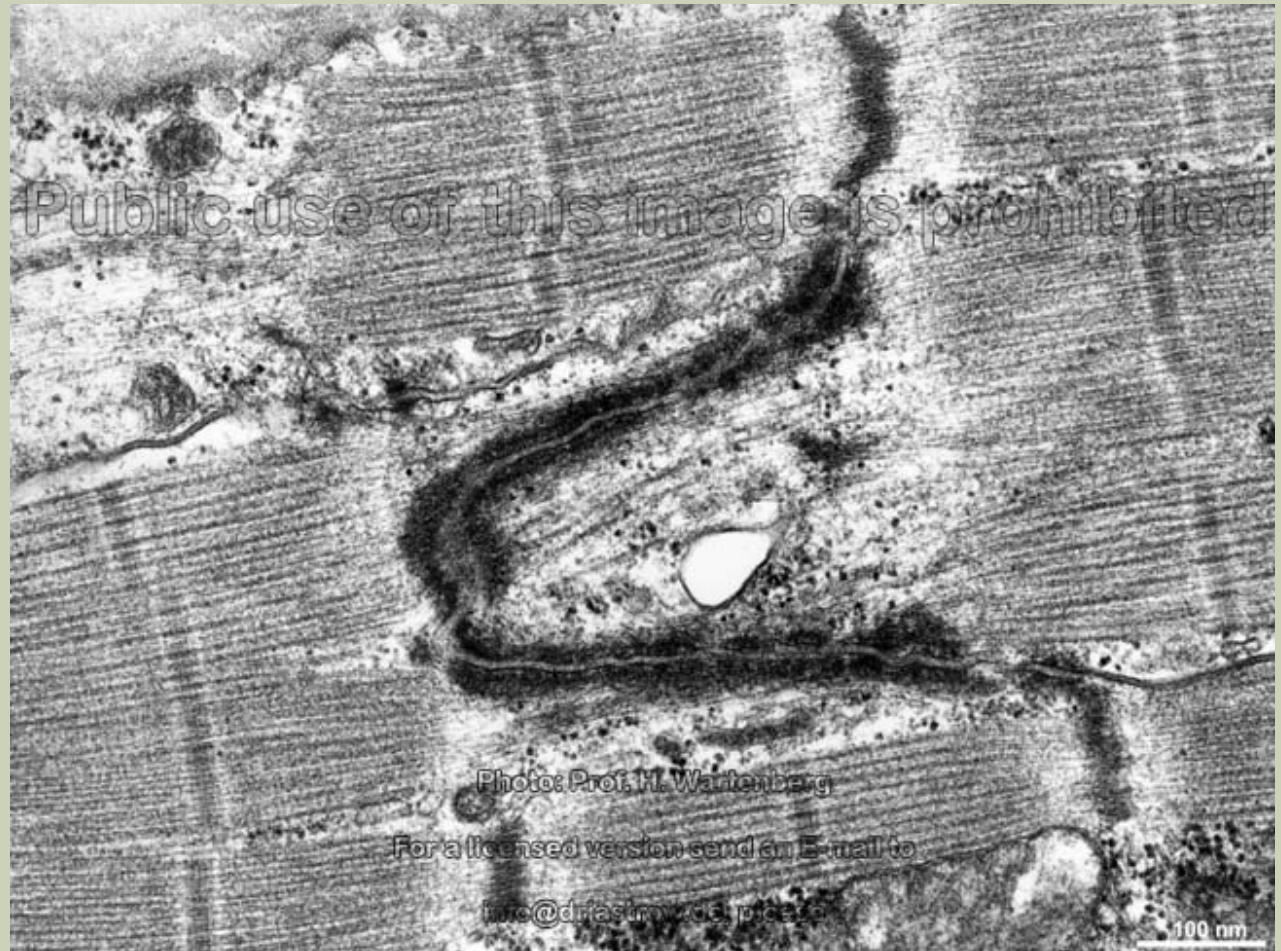
- The cell adhesion proteins of the desmosome, desmoglein and desmocollin, are members of the cadherin family of cell adhesion molecules
- They are transmembrane proteins that bridge the space between adjacent cells by way of homophilic binding of their extracellular domains to other desmosomal cadherins on adjacent cell.
- The extracellular domain of the desmosome is called the extracellular core domain (ECD) or the desmoglea, and is bisected by an electron-dense midline where the desmoglein and desmocollin proteins bind to each other.
- On the cytoplasmic side of the plasma membrane, there are two dense structures called the outer and inner dense plaque. These are spanned by the desmoplakin protein. The outer dense plaque is where the cytoplasmic domains of the cadherins attach to desmoplakin via plakoglobin and plakophilin. The Inner Dense Plaque is where desmoplakin attaches to the intermediate filaments of the cell.



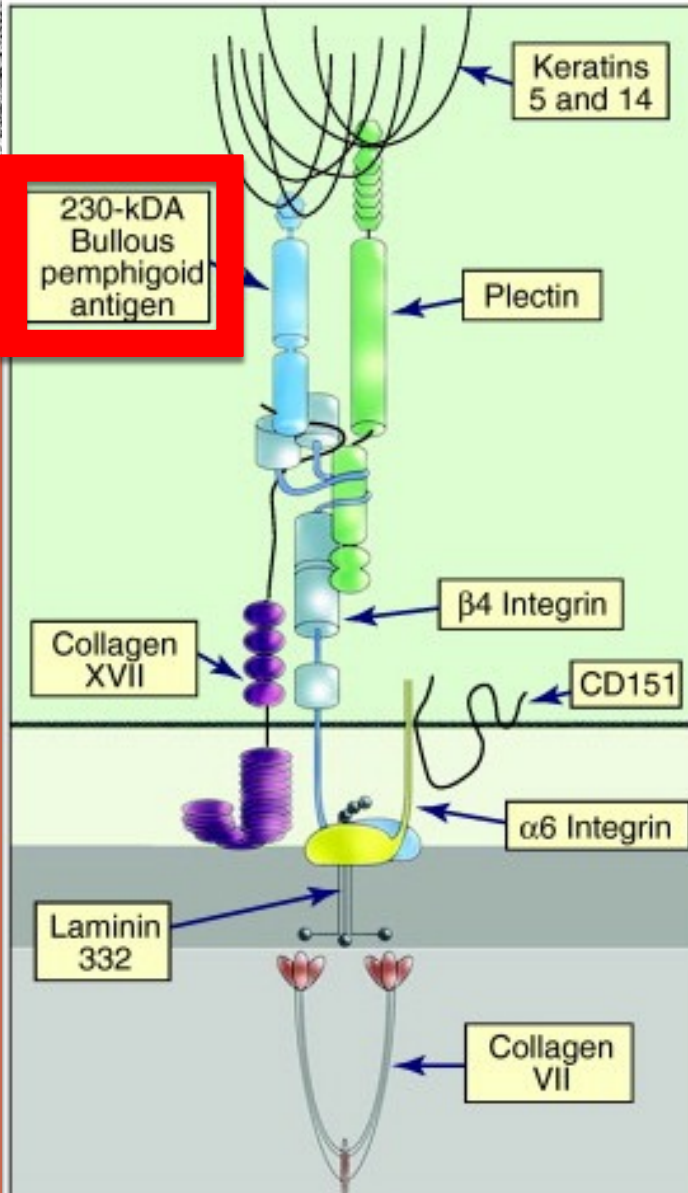
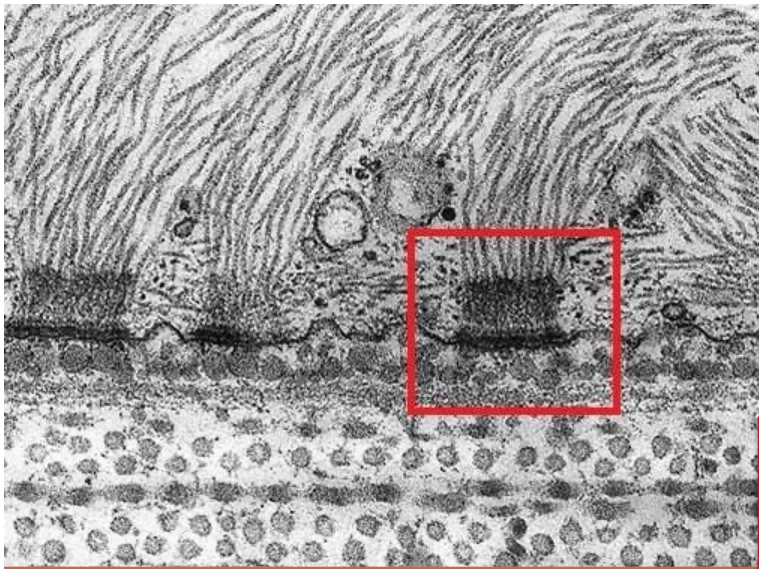


DESMOSOMES (MACULA ADHERENS)

- Distribution not limited to epithelial tissues.
- Desmosomes seen here in skeletal muscle



HEMI-DESMOSOME



Keratin 5: *KRT5*

EB simplex (AD)
Dowling-Degos disease (AD)

Keratin 14: *KRT14*

EB simplex (AD and AR)
Naegeli-Franceschetti-Jadassohn syndrome (AD)

Plectin: *PLEC1*

EB simplex-muscular dystrophy (AR)
EB simplex-pyloric atresia (AR)
EB simplex (AD)

230-kDa bullous pemphigoid antigen: *DST*

EB simplex (AR)

Collagen XVII: *COL17A1*

Non-Herlitz junctional EB (AR and AD)

α6 integrin: *ITGA6*

Junctional EB-pyloric atresia (AR)
EB simplex-pyloric atresia (AR)

β4 integrin: *ITGB4*

Junctional EB-pyloric atresia (AR)
Non-Herlitz junctional EB (AR)
EB simplex-pyloric atresia (AR)
EB simplex (AR)

Laminin-332: *LAMA3, LAMB3, LAMC2*

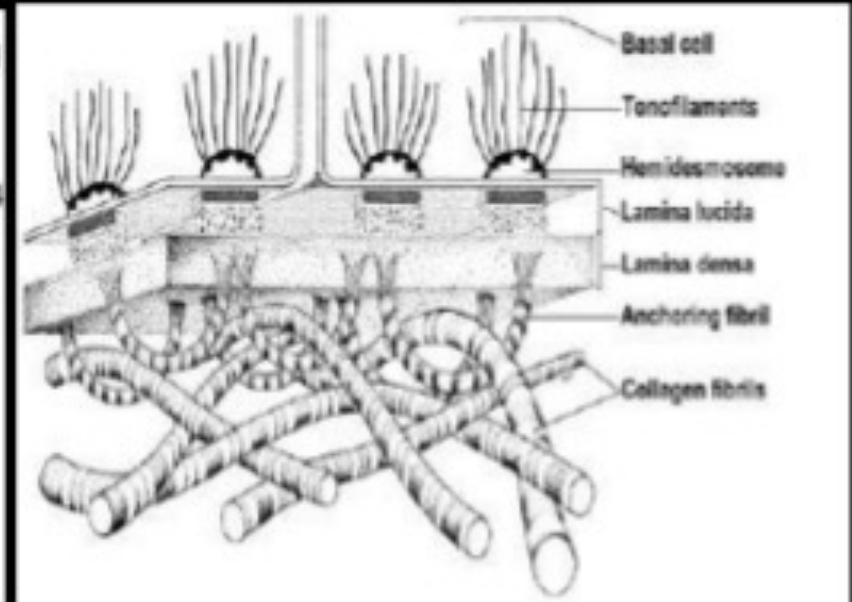
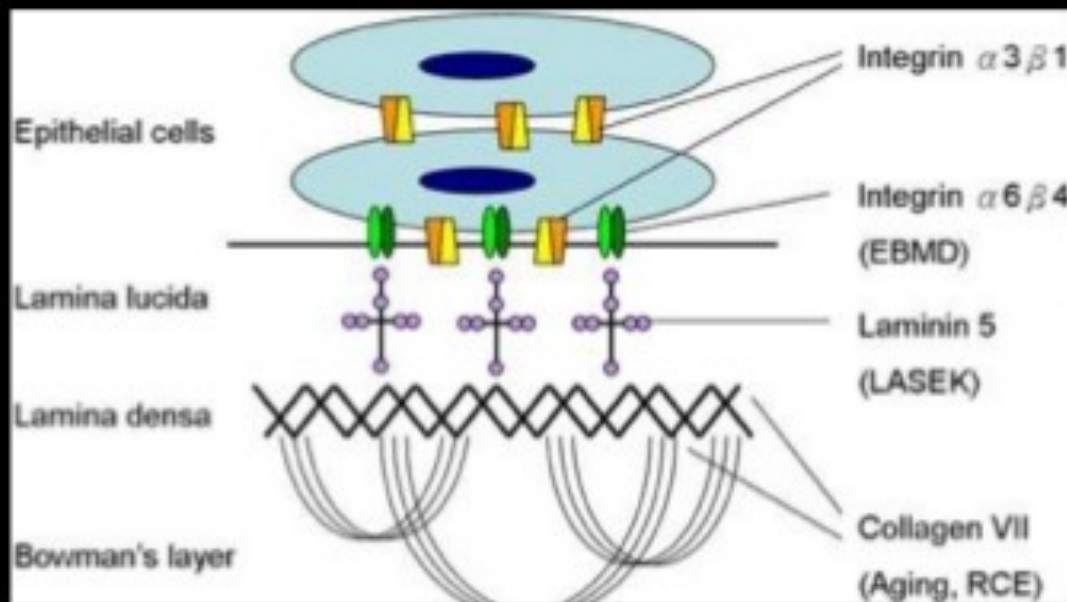
Herlitz junctional EB (AR)
Non-Herlitz junctional EB (AR)
Laryngo-oncho-cutaneous syndrome (*LAMA3*, AR)

Collagen VII: *COL7A1*

Dystrophic EB (AD and AR)

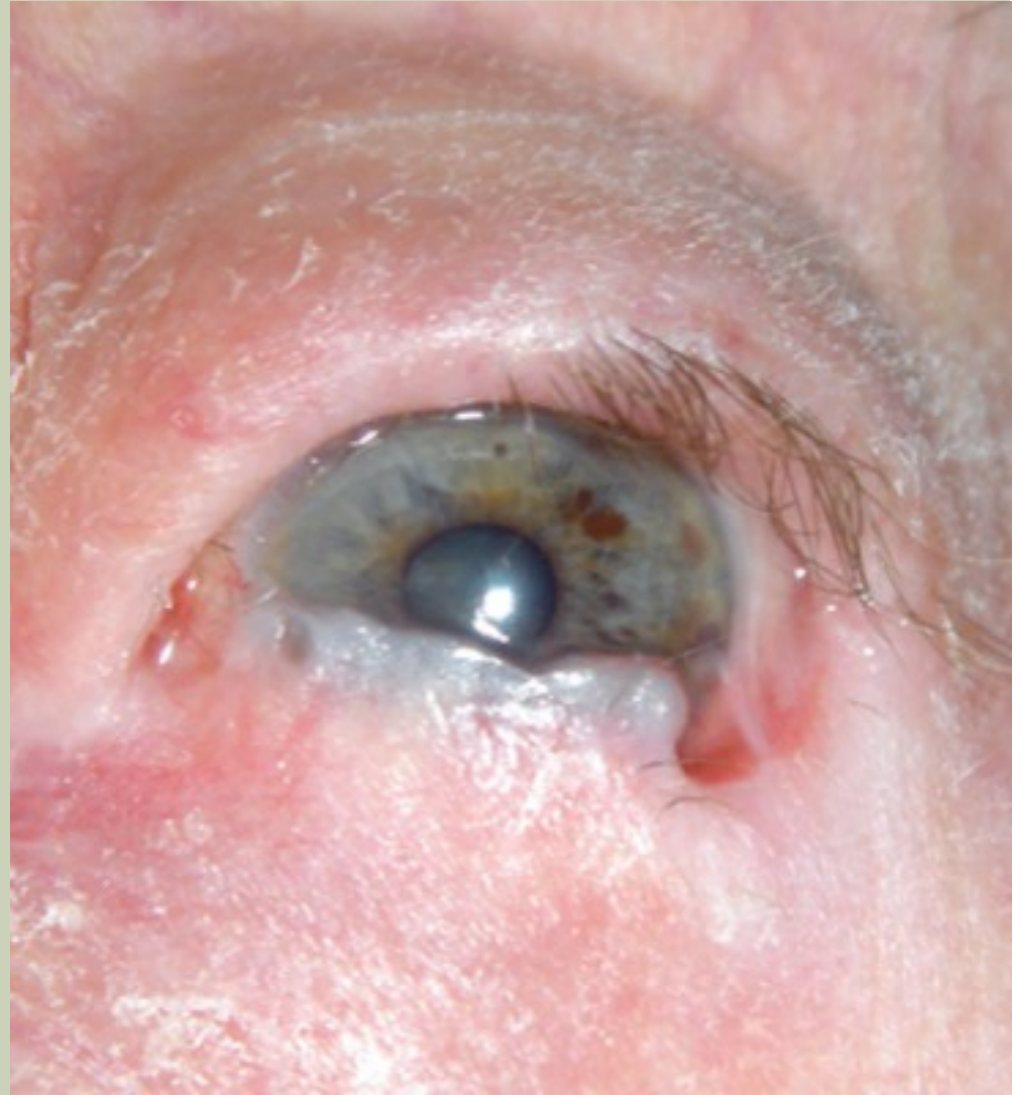
Basal cells are attached to basal lamina by hemidesmosomes.

Epithelial cell-cell contact is made through desmosomes - macula adherens. These are anchored intracellularly by tonofibrils.



OCULAR CICATRICAL PEMPHIGOID

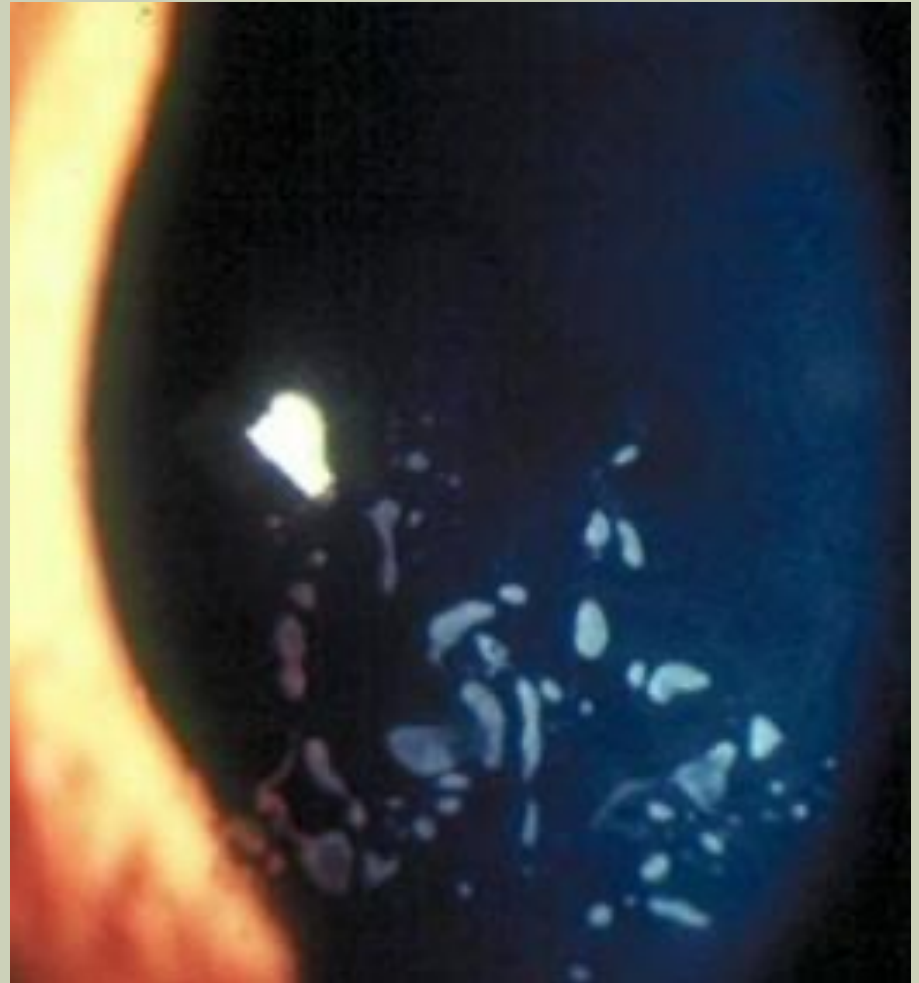
- Mucous membrane pemphigoid (MMP) is a rare, inflammatory autoimmune disorder characterized by blistering lesions that affect the mucous membranes of the body, especially the mouth and the eyes.
- Typical symptoms include redness, pain, tearing, and sensitivity to light. The inflammatory lesions on the ocular surface may result in scarring, loss of tear film, adhesions of the lids to the eyeball, corneal ulcers, and perforation.
- The frequent scarring that often occurs in this condition can lead to severe vision loss.



EPITHELIAL BASEMENT MEMBRANE DISEASE

Improper development of integrins leads to epithelial basement membrane disease, also known as fingerprint or map-dot dystrophy of the cornea.

- **With poorly adherent epithelium, these patients are prone to episodes of corneal epithelial erosion with pain and risk of infection.**



RECURRENT CORNEAL EROSION

- There is a history of previous corneal injury and abrasion, especially if caused by paper or fingernail
- Hemidesmosomes are not properly repaired, leading to recurring episodes of abrasion that often occur with the initial opening of the eyes in the morning.

